

Sentiment Analysis Methods: Survey and Evaluation

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Abstract

This paper establishes a rigorous framework for sentiment analysis applications in accounting and finance and demonstrates how to practically deploy and evaluate Natural Language Processing (NLP) models from lexicon-based approaches, to machine learning, and the latest powerful Bidirectional Encoder Representations from Transformers (BERT) models. We conduct an empirical evaluation of the performance of 31 different sentiment analysis methods based on a sample of 20,000 Glassdoor employee reviews, using a variety of different metrics and across seven different experimental settings. We find that, while the lexicon-based approaches remain the most popular with researchers, the performance of sentiment analysis can be doubled or even tripled with more advanced models. The experiments also show that all models perform best when assessing the sentiment of positive or negative statements or distinguishing between one- and five-star reviews, but more caution must be employed when including more ambiguous cases (two-star and four-star reviews) or using a multi-class sentiment division (positive, negative or neutral) rather than binary. The majority of the previous analysis of sentiment from Glassdoor reviews has focused on the star ratings but our findings suggest that analysis of the sentiment of text comments may add further nuance.

Keywords: sentiment analysis; textual analysis; lexicon-based approach; machine learning; BERT; employee reviews

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1 Introduction

The rise of digital platforms means that an unprecedented amount of textual data from news and social media is now easily accessible to academics and market participants. While the information within the textual data is abundant and potentially valuable to gain insights into public opinions and predict future trends, at the same time, it can be noisy and the sheer volume of material overwhelming. Therefore, researchers need to select the most efficient tools to identify, extract and analyse the most relevant information (Kobayashi et al. 2018, Pandey and Pandey 2019, Abbasi et al. 2019, Hickman et al. 2022, Bochkay et al. 2022). In this paper, we focus on an emerging topic in accounting and finance, sentiment analysis, a task that allows researchers to process the massive information flow by computationally identifying the emotions behind textual data.

As one of the most popular datasets used in accounting and finance research, Glassdoor stores over 70 million employee reviews, in both structured (numerical ratings) and unstructured (text comments) formats, relating to more than 600,000 companies worldwide. However, most empirical analysis focuses more on the numerical star rating (from one to five) provided by the reviewers as a measure of sentiment rather than considering the content of any text comments. Using the overall star ratings, researchers have demonstrated that employee satisfaction can be associated with ROA, Tobin's Q and operating margin to predict corporate performance (Melián-González et al. 2015, Huang et al. 2015, Symitsi et al. 2018, Stamolampros et al. 2019, Corritore et al. 2020). In addition, changes in employee satisfaction and flexibility reflected in their star ratings can also impact on stock returns (Huang et al. 2015, Au et al. 2021), customer contact (Wolter et al. 2019), and business outlooks such as future operating performance (Huang et al. 2020) and corporate disclosure (Hales et al. 2018).

Previous studies using Glassdoor text comments have employed text mining techniques such as topic modelling (Schmiedel et al. 2019, Symitsi et al. 2021), dictionary-based text analysis programmes such as DICTION, Linguistic Inquiry and Word Count (LIWC) and WordNet (Symitsi et al. 2018, Stamolampros et al. 2019, Corritore et al. 2020), and data-mining software like IBM Watson (Dabirian et al. 2017, 2019) to extract

factors that affect employee satisfaction including: organizational structure (Huang et al. 2015, Creek et al. 2019), culture (Robertson et al. 2019, Canning et al. 2020), financials (Jing et al. 2019) and policies (Storer and Reich 2021). Moreover, Tambe et al. (2020) use cluster analysis of text from Glassdoor reviews to claim information technology (IT) workers prefer to work for companies that use emerging technologies in part because they value technology and learning on the job. Campbell and Shang (2022) use an inverse regression approach to derive importance weights for words and demonstrate that attributes related to corporate misconduct are widespread among employee comments. Sull et al. (2022) identify 172 topics frequently mentioned in Glassdoor employee reviews and study the sentiment related to each topic, finding that cultural toxicity drives up employee turnover.

We find the methods used in the textual analysis of employee evaluations are still relatively homogeneous. While Bochkay et al. (2022)'s work provides a guideline for implementing Natural Language Processing (NLP) models in accounting and finance, our work is specialised in sentiment analysis and presents the first rigorous and comprehensive comparison of 31 different sentiment analysis methods using the text from 20,000 Glassdoor employee reviews. The methods we adopt in this paper can be divided into three categories: the lexicon-based approaches (i.e., Harvard IV-4 dictionary (HIV4), Loughran-McDonald (LM) dictionary, SentiWordNet, and VADER); the machine learning approaches (i.e., Logistic Regression (LR), Naive Bayes (NB), Support Vector Machines (SVM), Decision Tree (DT), Random Forests (RF) and Extreme Gradient Boosting (XGB)); and the pre-trained models (i.e., Bidirectional Encoder Representations from Transformers (BERT) and its extensions). By comparing these methods, we demonstrate how NLP models can be applied to sentiment analysis, discuss the trade-offs between models, and how to select a suitable metric to evaluate the performance of the models.

Our work also expands on this literature by conducting a comprehensive review of sentiment analysis methods previously applied in accounting and finance studies and developing a framework that summarises current sentiment analysis methods based on

the fundamental characteristics of texts and the types of NLP tools used for processing them. A recent study by Huang et al. (2022) found that less accurate sentiment measures tend to underestimate the economic impact of the sentiment of analyst reports and, as a result, have less explanatory power compared to the models with higher accuracy. Their paper utilises manually labelled analysts' reports as their sample for testing sentiment predictions. This has several limitations. Firstly, their sample size is smaller at only 10,000 sentences. Secondly, there may be subjectivity bias from the researcher labelling the sentence. In contrast, our approach using Glassdoor avoids the need for manual labelling, allowing for larger samples and reducing the potential for researcher bias in labelling. Additionally, the sentences in Huang et al. (2022) are not evenly distributed among the sentiment categories, and they do not use the Matthews Correlation Coefficient (MCC) as their metric. This could potentially lead to issues with the other metrics used in their analysis. Furthermore, Huang et al. (2022) only considers the sentiment of individual sentences, whereas our approach takes into account the sentiment of the whole review, which consists of at least three sentences (pros, cons, and advice).

We design seven different experiments using both the text and star ratings from Glassdoor reviews to investigate the extent to which the sentiment identified from the text matches the numerical rating. Our preferred evaluation metric is the MCC score, but we also apply four other standard metrics (Precision, Recall, F1-Score, and Accuracy) to ensure the robustness of our results. Our findings clearly show that the BERT models consistently deliver the best performance in all experiments, followed by the machine learning approaches with word embeddings and lastly the lexicon-based approaches. At the same time, we can also observe that the performance of all different approaches declines as the ambiguity of the text increases. For instance, all models perform best in Experiment 1, which simply tests whether they identify text comments as positive or negative, and worst in Experiment 7, which introduces a neutral sentiment class as well as positive or negative. This demonstrates that researchers must use caution when employing automated textual sentiment analysis in particularly complex or ambiguous cases. In addition, Xu et al. (2021) have suggested that contextual descriptions can

actually provide a better indication of users' perceptions of quality than their numerical ratings. We also find some discrepancies between a review's sentiment as indicated by its overall star rating and by the text comments. Numerical ratings are good indicators of the overall sentiment but it is likely that valuable additional information can be extracted from the associated text comments.

The structure of this paper is as follows. After this introduction, Section 2 introduces the framework for the sentiment analysis methods most used in accounting and finance studies and reviews the related literature in detail. We distinguish between conceptual approaches involving human judgement, hybrid approaches based on the text contents, and empirical approaches to assign sentiment such as reaction studies. The hybrid approaches can be divided into deductive, mostly lexicon-based which we discuss first, and inductive approaches (machine learning or pre-trained models) that use word embeddings and machine learning classifiers. Section 3 first describes and compares the word embeddings. These are the processes by which text is transformed into vectors for automated sentiment classification. We also discuss several popular machine learning classifiers that are used in combination with the above word embeddings for sentiment classification in Section 4. This is followed by the pre-trained models (the BERT-related models) in Section 5. After setting out the background and key features of the different sentiment analysis methods, we move on to our empirical assessment of their performance. We introduce the Glassdoor data and explain the experimental setup as well as the evaluation metrics and result discussion in Section 6. Finally, Section 7 summarises our key findings and lays out several possible directions for further research.

2 Sentiment Analysis Framework

There are a wide variety of sentiment analysis techniques that are based on different approaches and have different levels of complexity. This can make it difficult for researchers to identify the best approach to use in their research. Based on a systematic review of the current literature in accounting and finance and partly considering the amount of hu-

man intervention with the textual data, we have categorised these techniques into three broad categories shown in Figure 1: 1) *Pure Conceptual Approach*; 2) *Hybrid Approach*; 3) *Pure Empirical Approach*. Further details of the literature we reviewed to develop this framework are outlined in Appendix 1.

[INSERT Figure 1 ABOUT HERE]

Pure Conceptual Approach

In the Pure Conceptual Approach, the sentiment of a text is solely determined by human perception. The perception of this sentiment can be formed during or assigned after the text is generated, and by the creator of the text or a later reader. In most cases, a researcher has to rely on manual annotation to assign sentiment to a text. This process is usually applied to existing text that has not already been labelled for its sentiment. Manual annotation is frequently used to provide a training sample for sentiment analysis models. When dealing with a large unlabelled corpus of text, for example, it is common to label a sample from this corpus manually, and use this sample to train a model as a classifier so that it can be applied to the rest of the corpus. Using manually annotated text, Antweiler and Frank (2004) find the tone of stock messages on the internet help predict market volatility, and the effect on stock returns is statistically significant but economically small, Li (2010) suggests the tone of forward-looking financial statements is positively associated with future earnings, Huang et al. (2014) discover investors react more strongly to negative texts than to positive ones, and Abbasi et al. (2019) show that user-generated content on social media can detect adverse events in advance, and false-positive rates are further reduced after including negative sentiment polarity in the models. In addition, this approach is also used to create standard and high-quality datasets for the NLP community to test new models.

One fundamental concern with this approach is that it is inherently subjective and different readers might interpret the same text differently, particularly in more ambiguous cases. To combat this, in the Financial Phrase Bank (Malo et al. 2014), a dataset consisting of around 5,000 sentences from LexisNexis categorised by sentiment, a piece of text

is usually evaluated by multiple annotators to reduce human factor biases and concluded by the strength of majority agreement. Araci (2019), Yang et al. (2020), Huang et al. (2022) used this dataset to extend the BERT model with finance-specific domain knowledge, known as FinBERT. Another issue is the time-consuming and resource-intensive process of manually annotating unlabelled text, particularly if this is being carried out by multiple different annotators to reduce bias.

Some sources, however, have already been labelled with their sentiment. For instance, during the Glassdoor review process, the reviewer first assigns a numerical star rating to the company and then provides text comments highlighting pros and cons with an option to provide further information such as “advice to management”. Another example is Stocktwits, which includes a feature for users to tag their messages as either “bullish” or “bearish” as a way of expressing their sentiment in addition to the message’s textual content. In such self-reporting systems, an indication of the sentiment is provided by the original creator of the content. In this paper, we will use the self-reported star ratings in Glassdoor reviews as our “ground truth” for the intended sentiment of the review and test how accurately the different sentiment analysis methods can identify this sentiment from the associated text comments. Our methodology will be explained in more detail below but, for now, we may note that it has two key advantages: first, it removes the possibility that a subjective human annotator might assign a different sentiment to that intended by the creator; and second, it reduces the resources needed for manual annotation and enables a larger sample to be used.

Hybrid Approach

The Hybrid Approach consists of two sub-categories: the Deductive Encoding of Meaning and the Inductive Encoding of Meaning. The former uses a pre-existing set of rules to deduce the sentiment of the text and is associated with lexicon-based methods of sentiment analysis, while the latter aims at analysing the text to develop a set of rules that can assign a sentiment to the text and is linked to machine learning and transformer models. This could be seen as a hybrid approach as it requires a certain level of text processing

but not the same degree of manual annotation as the Pure Conceptual Approach.

Deductive Encoding of Meaning

The Deductive Encoding of Meaning applies an established framework to deduce the meaning of a text. In the field of sentiment analysis, this method uses pre-defined rules to score the sentiment of words in a document and generates an aggregated value for the polarity between positive and negative words. In the introduction, we listed several examples of rule-based programmes and software such as DICTION, LIWC and IBM Watson. Here we focus on the lexicon-based approaches that rely on dictionaries or word lists. Lexicon-based sentiment extraction is one of the most popular approaches used on financial text, not only for its relatively straightforward application but also because it does not require labelled text, and most financial sources are not labelled.

One of the earliest lexicons, General Inquirer (GI), was developed at Harvard University in the 1960s for automated content analysis and attempted to tag words across 182 categories with dictionaries (Stone and Hunt 1968). The Harvard IV-4 dictionary (HIV4) of GI assigns a positive and a negative semantic dimension to words and is often used for sentiment analysis. Tetlock (2007) use HIV4 to detect sentiment in news articles and find that high values of media pessimism are associated with downward pressure on market prices. Later, they also found that the sentiment of text can be used to predict individual firms' accounting earnings and stock returns (Tetlock et al. 2008). Engelberg et al. (2012) suggests that negative news will increase the negative relation between short sales and future returns. Using the same approach, some research has discovered that a negative tone in the text of 10-Ks and 10-Qs can increase the cost of capital and return volatility (Kothari et al. 2009). Feldman et al. (2010) identify that changes in sentiment in the management discussion and analysis (MD&A) section of a firm's SEC filing are a significant predictor of short-term market reaction. Using HIV4, Yukselturk and Tucker (2015) reveal varying relationships between themed sentiments and analyst outputs across pre-crisis, crisis, and post-crisis periods. Yekini et al. (2016) show that the tone of narratives published by UK companies in annual reports is related to market

reaction, and should not be considered only as impression management tools but also as a way of disseminating price-sensitive information.

The creation of a dictionary also benefits from domain-specific knowledge. While the HIV4 dictionary has been proven efficient on general texts, the Loughran-McDonald (LM) dictionary offers more specific insights into financial texts. Loughran and McDonald (2011) examined a large sample of 10-K filings between 1994 and 2008 and concluded that nearly three-quarters of negative words according to the HIV4 dictionary appeared to be *non-negative* in an accounting and finance context, and so they created the LM dictionary by evaluating the co-occurrence and frequency of words in the corpus. Both HIV4 and LM dictionaries interpret polarity employing a Lydia system (Godbole et al. 2007).

$$Polarity = \frac{(Pos - Neg)}{(Pos + Neg)} \quad (1)$$

Pos and *Neg* are word counts for the words in positive and negative sets. $Polarity > 0$ implies a positive sentiment whereas $Polarity < 0$ suggests a negative sentiment for the text. Rogers et al. (2011) show firms that have more optimistic statements in their earnings announcements suffer from higher litigation risk. Price et al. (2012) suggests both positive and negative tones of conference calls are significantly associated with abnormal returns and trading volume. Similarly, Tsai and Wang (2017) report that finance-specific sentiment lexicon has a strong correlation to financial risks, the greater the amount of finance-specific sentiment, the higher the risk. From a different perspective, Feuerriegel and Gordon (2019) finds that high-dimensionality text input from financial news can lead to over-fitting in machine learning models used to predict macroeconomic indicators, therefore, the LM dictionary can be a better solution.

Another lexicon-based sentiment analysis approach is SentiWordNet (Esuli and Sebastiani 2006, Baccianella et al. 2010), an extension of WordNet which was introduced by (Miller 1995) to expand opinion mining by evaluating words' semantic relations based on synsets. A synset is a group of synonyms and antonyms that share the same context, and each synset expresses a distinct concept but is interlinked by their conceptual relations. SentiWordNet assigns a numerical score between 0 and 1 for positivity, negativity or ob-

jectivity to each synset. Any one of these three scores can be 0 as long as they sum up to one because the scores suggest a proportional agreement of the sentiment depending on the context. The polarity is calculated as the difference between positive and negative scores of each synset s .

$$Polarity = Pos(s) - Neg(s) \quad (2)$$

This can raise a problem in some cases where synsets have no positive or negative polarity score. Moreover, the scores are generated through a semi-supervised step followed by a random walk step, and therefore cause a lot of noise in the SentiWordNet lexicon compared to the human-validated dictionaries.

In 2014, VADER (Valence Aware Dictionary for Sentiment Reasoning) was created to overcome some of the challenges that SentiWordNet had encountered with social media textual data (Hutto and Gilbert 2014). VADER interfaces with SentiWordNet and adapts the difference between positive and negative scores of each synset as valence to differentiate a word's sentiment intensity. For instance, the words "good" and "excellent" both express positive sentiments, but "excellent" is more positive thus VADER will give it a higher sentiment rating. In addition to modifying the degree of intensity, VADER also incorporates the impact of punctuation, capitalisation, contrastive conjunction and negation. VADER was developed by obtaining a large number of ratings for words in the existing lexicons using crowd-sourced resources from Amazon's Mechanical Turk. After assigning a rating to each word, VADER aggregates the ratings of all words in a text and produces a positive, a neutral, and a negative score normalised between -1 and 1 to indicate the proportion of text that falls in these categories. VADER has shown impressive results in many studies, and it also has the advantage of supporting emoticons and acronyms for sentiment analysis.

Despite a large amount of contextual interpretation and human-factored decision-making required in the development of these dictionaries, the word lists within the dictionaries can be relatively limited as well, which means that the dictionaries are not able to assign a sentiment score to words that are outside of the pre-defined lists.

Inductive Encoding of Meaning

On the other hand, the Inductive Encoding of Meaning develops an analytical mechanism derived from the data itself. This can be done using machine learning or pre-trained models.

van der Heijden (2022) examines the use of machine learning in accounting research and applies it to predict a firm's industry sector using publicly available financial statement data. The results show that machine learning algorithms can accurately predict industry sectors, and can be valuable in accounting domains where prediction is the main focus. Commonly, texts need to be represented numerically before being inserted into training, which can be done through bag-of-words (BOW) or term frequency-inverse document frequency (TF-IDF), these word embeddings will be examined in more detail in the following Section 3. Previous studies have reported that these text representations combined with machine learning models, such as NB, LR, SVM, DT, RF and Maximum Entropy (MaxEnt), performed very well on financial news and tweets (Antweiler and Frank 2004, Hagenau et al. 2013, Nassirtoussi et al. 2014, Abbasi et al. 2019, Renault 2020, Frankel et al. 2022). These machine learning classifiers will be examined in more detail in Section 4 below. Li (2010) and Huang et al. (2014) have compared the NB algorithm with dictionary-based approaches to classify the sentiment of analyst reports and have shown NB algorithm achieves better performance. In contrast, Sohangir et al. (2018) concluded that VADER outperforms LR, NB and SVM in extracting sentiment from financial social media. In our evaluation of these methods, we included more extensive analysis to test which approach performs better, we found the performance of machine learning rely on the choice of word embedding and hyperparameters.

Most recently, the introduction of the BERT model (Devlin et al. 2018), a state-of-the-art language representation model in NLP, has increased the accuracy of sentiment analysis to a new level. This approach lies between deductive encoding and machine learning because it inherits the features from pre-training but also relies on fine-tuning sample data for downstream tasks. Sousa et al. (2019) have conducted experiments using financial news to show that BERT (82.5% in accuracy) outperformed NB and SVM

(69% in accuracy). Further studies using different textual data have also reached the same conclusion that BERT achieved the best accuracy compared to other lexicon-based and machine learning approaches (González-Carvajal and Garrido-Merchán 2020, Mishev et al. 2020, Zhao et al. 2021, Stevenson et al. 2021). In addition, studies have proposed different versions of FinBERT to gain more domain knowledge in accounting and finance (Araci 2019, Yang et al. 2020, Liu et al. 2021, Huang et al. 2022). Another study, Bingler et al. (2022) introduce ClimateBert which specialises in climate-related texts. Although, as Kriebel and Stitz (2021) point out simple and complex deep learning architectures both provide comparably good results. As these are the most complicated methods, we examine them further in Section 5 below.

Pure Empirical Approach

The last class of sentiment analysis is the Pure Empirical Approach. This does not rely on human judgement or deducing/inducing sentiment from the textual data itself. Instead, the Pure Empirical Approach “let the market” decide the sentiment reflected in the text. For instance, in reaction studies the sentiment of financial reports, news and tweets is determined by observing changes in stock prices (Hagenau et al. 2013, Jaggi et al. 2021, Frankel et al. 2022). Alternatively, it has been proposed that default records could be used as a measure of the sentiment of credit reports. Kriebel and Stitz (2021) claim that textual information can improve credit default predictions. Conversely, Stevenson et al. (2021) argue that textual loan information produces relatively accurate predictions alone, however, it does not offer additional performance improvement when used with structured data. There are limitations to this approach. It is not always possible to make a clear link between a text and a market reaction; it would be presuming too much to attribute an increase (decrease) in the market value of a large firm to a single positive (negative) employee review on Glassdoor. It also limits the analytical usefulness of the identified sentiment. If a text has been identified as having positive sentiment based on a subsequent increase in firm value, then it would be tautologous to use that sentiment as an explanatory factor for firm performance. The Pure Empirical approach could be

useful to assign sentiment labels to a sample of text that could be used to train a machine learning model and then applied to a larger corpus. We do not use this approach in the paper since we can use the user-assigned star rating in Glassdoor reviews as a proxy for intended sentiment.

3 Word Embeddings

We will explore various approaches in our analysis, previously we discussed lexicon-based approaches, in the following sections we will explain machine learning and BERT related approaches in more detail. However, in order to use the more advanced methods for sentiment classification, the first and most important thing is to turn unstructured text data into a structured format. Word embeddings are techniques designed to represent a text by a string of numbers. The numbers are called vectors and this process of this transformation is known as text vectorization.

Bag-of-Words

The bag-of-words (BOW) model is the simplest feature representation method that vectorizes text. As its name suggests, the model forms a vocabulary of all the unique words in a document and counts the occurrence of each word. However, the model ignores the order of words in a document, such that contextual information will be lost without knowing where the words occurred. In addition, the size of the vocabulary will be very large if new words occur in a new document. Increased vector lengths raise an issue of a sparse matrix where most elements in the vectors are zeros.

Term Frequency-Inverse Document Frequency

Whilst the BOW model is simple to use, it is biased towards words with high occurrence. High-frequency words such as “and” and “the” may not contain relevant or meaningful information about the documents. Term Frequency-Inverse Document Frequency (TF-IDF) overcome this problem by re-scaling the word frequency across all documents and

providing a weight for each word. The weighting is achieved by multiplying two matrices: a term frequency matrix $tf(t, d)$ measuring the word frequency in a document, and an inverse document frequency matrix $idf(t, D)$ capturing the distinctiveness of a word across the entire corpus. As a result, words containing important information will be assigned with a relatively high TF-IDF weight.

$$tf(t, d) = \frac{f_d(t)}{\max_{w \in d} f_d(w)} \quad (3)$$

$$idf(t, D) = \log\left(\frac{|D|}{|\{d \in D : t \in d\}|}\right) \quad (4)$$

$$tfidf(t, d, D) = tf(t, d) \times idf(t, D) \quad (5)$$

$f_d(t)$ is the frequency of term t appears in the document d ; $\max_{w \in d} f_d(w)$ is total number of terms in the document; ϵD is the corpus of documents; $\{d \in D : t \in d\}$ is the number of documents with term t .

Word2Vec

A fascinating improvement made on word embeddings was when Mikolov et al. (2013) from Google introduced the Word2Vec model that takes advantage of the neural network structure. In the previous BOW and TF-IDF models, vectors used to represent text are only numbers without any semantic meanings, and the vector space is proportional depending on the vocabulary size. The Word2Vec model generates a unique vector for each word in the document, and instead of storing all the information of the vocabulary, the idea is to design a model that groups the vectors of similar words in the vector space. Word2Vec has two architectures to achieve this, a continuous bag of words model (CBOW) and a continuous Skip-Gram model, as shown in Figure 2.

[INSERT Figure 2 ABOUT HERE]

The CBOW model aims to predict the centre word by its surrounding words with a fixed window size, while the Skip-Gram model does exactly the opposite, it is trained to predict the probability of the surrounding neighbour words given a centre word. Since the

Skip-Gram model takes single words of input, it is less sensitive to frequent words and it has a better representation of rare words, whereas CBOW tends to overfit high-frequency words. However, Skip-Gram is computationally more expensive because it needs to predict several outputs for the context words whilst CBOW is only trained to predict one centre word at a time. In general, the Word2Vec model has the advantage of maintaining the semantic relations of different words in a document with little requirement of computer memory, and these encoded properties can be easily extended for sentiment analysis. One of the disadvantages of Word2Vec is that it handles out-of-vocabulary (OOV) words poorly because it assigns random vector representations for them, and the information retrieved is only locally optimal because a word representation relies on its neighbours.

Global Vectors for Word Representation

GloVe is short for Global Vectors for Word Representation, created by researchers from Stanford University (Pennington et al. 2014), and is an unsupervised learning algorithm that derives word embeddings through global word-word co-occurrence statistics. It combines a global matrix factorisation (e.g. latent semantic analysis (LSA)) and a local context window (e.g. Skip-Gram) in the training process. GloVe and Word2Vec are both unsupervised methods for word embeddings fitted up to 300-dimensional word vectors. Similar to Word2Vec, GloVe uses contextual information for text representation but the main difference between them comes from their structures of generating word vectors. Word2Vec is a predictive model that focuses on local context information of words, whereas GloVe is a count-based model which incorporates global statistics through dimensionality reduction on the co-occurrence counts matrix to obtain word vectors. Using different training methods, the Word2Vec embeddings provided by the Python library Gensim are pre-trained on Google News dataset for 3 million words and phrases ¹, while the GloVe embeddings introduced by Stanford are pre-trained on Wikipedia, Common Crawl, and Twitter with varying sizes ². In various NLP tasks such as analogy, word

¹<https://radimrehurek.com/gensim/models/word2vec.html>

²<https://nlp.stanford.edu/projects/glove/>

similarity and named entity recognition (NER), GloVe has demonstrated an outstanding performance compared to other models even with a small text corpus. The pre-trained word vectors of GloVe for large corpora are publicly available for researchers to access.

4 Machine Learning Classifiers

Sentiment analysis is essentially a classification task, where a classifier builds and trains on extracted features from text. We have discussed the word embeddings for feature extraction in the previous section, here we consider several supervised machine learning classifiers that are widely used for sentiment analysis and proven to be powerful in many studies. We are interested in comparing their performance in combination with different feature representation and word embedding techniques for sentiment analysis on our Glassdoor employee reviews.

Logistic Regression (LR):

LR is one of the most popular algorithms for its simplicity to implement and interpret. It operates with a hypothesis and a sigmoid function to determine the probability of an output sentiment. The range of the sigmoid function is between 0 and 1, with a threshold of 0.5, the sigmoid function transforms any value into a range from 0 to 1. Hence any probabilities that are greater than the threshold will lead to the class value of 1 (e.g. positive sentiment) otherwise 0 (e.g. negative sentiment). Logistic regression is usually used for binary classification, it assumes the features and classes are independent of each other and predicts a binomial probability. In multi-class sentiment analysis, standard LR can be modified by predicting a multinomial probability for the input features and changing the loss function from the Least Squared Error to Cross Entropy.

Naive Bayes (NB):

In the sentiment analysis, NB is also referred to as Multinomial Naive Bayes classifier (Rish et al. 2001). It uses the Bayes Theorem that predicts the probabilities of sentiment

class by using the joint probabilities of words and classes. With the assumption of feature independence, given the features $X = x_i$, and the sentiment classes $Y = y_i$.

$$P(Y|X) = \frac{P(Y) \cdot P(X|Y)}{P(X)} \quad (6)$$

Where $P(Y)$ is the prior probability and $P(X|Y)$ is the likelihood probability. It is fast computing compared to other classifiers, requires little training data and performs well in multi-class classification tasks. The main limitation of NB is that it assumes all the attributes are mutually independent implying no link between one word to another, which is rarely the case in real life.

Support Vector Machines (SVM):

SVM performs sentiment classification by assigning a hyperplane that best separates the positive and negative classes (Cortes and Vapnik 1995). The optimal hyperplane is found by maximising the distance between the hyperplane and each class. The data points used to define the hyperplane are referred to as the support vectors. In a binary sentiment classification, the classes are two-dimensional and the hyperplane can be as simple as a line, however, it will become more complex and non-linear if the dimension increases. Thanks to this attribute, SVM allows for higher accuracy when multi-dimensional features occur.

Decision Trees (DT):

Proposed by Quinlan (1986), this algorithm is built in the structure of a tree with decision nodes, branches and leaves as its name suggests. Each decision node represents a filtering criterion, and the branch shows whether the criteria are met then leads to an outcome of a class label on a leaf node. Starting from the root node, predictions on the leaves are made by continuously going through the nodes on the tree until the decision nodes can not be split further. DT algorithm works well on both numerical and categorical data and requires little data preparation. However, due to their hierarchical structure,

decision trees often over-fits the data and suffers from high variance issue where any small changes can cause instability of the predictions.

Random Forests (RF):

RF was first introduced by Breiman (2001) and functions by aggregating the predictions of a collection of DT. It adopts a type of ensemble method called bagging. With bagging, each decision tree is trained by a random subset of features independently, and their predictions are then averaged for an eventual prediction. The training can be efficiently done in parallel. Since RF selects the features randomly, it processes the inputs in a more generalised way without depending highly on any specific set of features and therefore usually outperforms DT.

Extreme Gradient Boosting (XGB):

Similar to RF, gradient boosting is an ensemble learner and a set of DTs as well. Different from bagging, boosting is another type of tree ensemble method. Instead of building each tree independently, gradient boosting builds one tree at a time and combines the predictions along the training process by constantly correcting the errors from the prior models. XGB is an implementation of the gradient boosting framework introduced by Chen and Guestrin (2016), and since its release, it has been one of the leading algorithms in Kaggle competitions and popular in sentiment analysis.

We will use all combinations of these word embeddings and machine learning classifiers in our study.

5 Pre-trained Models

Pre-trained Models are large neural networks that are used in a wide variety of NLP tasks. They operate under a “pre-train” to “fine-tune” paradigm: models are first pre-trained over a large scale of text and then fine-tuned on a specific downstream task.

The motivation behind pre-training is to understand the contextual meaning of a

word, and the word embeddings discussed previously ignore the order of words during their training. As a result, the vector of a word is static and always the same across any sentence or document even though the same word might have different meanings in a different context. By contrast, the contextual embedding such as BERT (Devlin et al. 2018) relies on a Transformer architecture that takes into account the sequence of all words and their positions in a sentence from both left-to-right and right-to-left contexts simultaneously.

Transformer

The Transformer is a state-of-the-art deep learning model in NLP for processing sequential data such as language modelling and machine translation (Vaswani et al. 2017). It adopts a self-attention mechanism that enables the model to transform one sequence to another with an encoder and a decoder and allows for more parallelization during training. Intuitively, attention is a function that relates different positions of the input sequence to compute a representation of that sequence. From each attention unit, the Transformer learns a set of Query, Key, and Value weight matrices denoted by W_Q , W_K , W_V respectively during the model training. The first step in calculating self-attention is to use the word embedding matrix X to multiply the weight matrices to produce three vectors; a query vector: $Q = XW_Q$, a key vector $K = XW_K$, and a value vector $V = XW_V$. The second step is to calculate the self-attention weights by taking the dot product of the query vector and the key vector, then dividing it by the square root of the dimension of key vector $\sqrt{d_k}$ for stabilising gradients. Thirdly, the attention weights are passed through a softmax to get normalised scores which are then multiplied by the value vectors by a compatibility function. Lastly, the output of the self-attention layer is computed as the sum of the weighted value vectors.

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (7)$$

Running through an attention mechanism several times in parallel, the multi-head attention in the Transformer model operates by mapping 8 pairs of queries, keys, and values. After concatenating and linearly transforming the independent attention outputs, the multi-head attention enables the model to jointly attend to information between words at various positions.

$$MultiHead(Q, K, V) = Concat(head_1, \dots, head_h)W^O \quad (8)$$

$$head_1 = Attention(QW_i^Q, KW_i^K, VW_i^V) \quad (9)$$

where W_i^Q , W_i^K , W_i^V and W_i^O are learnable parameter matrices.

Figure 3 shows a demonstration of the structure of the Transformer. The encoder (the left half of Figure 3) builds a continuous word embedding for sequential representation and feeds into the decoder (the right half of Figure 3) to generate an output sequence. The output embedding comes from the embedding layer which is shared with the input embedding, and the positional encoding is to make sure the model learns the order of the sequence. The encoder and decoder are composed of a stack of six identical layers, and each layer mainly consists of a feed-forward network and a multi-head attention module.

[INSERT Figure 3 ABOUT HERE]

In a feed-forward neural network, the sequential information is passed on in one direction, from the input layer through hidden layers and to the output layer. It is different from Recurrent Neural Networks (RNNs) where the information cycles through a loop, and every decision is made based on a current and a previous input. Without communications or inferences with the previous computation in each position, the Transformer model successfully reduces the training time. A sequence is processed as a whole instead of word by word, therefore, the Transformer model is not affected by long dependencies. In the decoder, masking is applied to the first sub-layer to ensure the predictions for a position are only dependent on the known outputs before such position. This is then connected with the outputs from the encoder and eventually passes through a softmax layer to generate a prediction of the output sequence.

Bidirectional Encoder Representations from Transformers

Building on the foundation of the Transformer architecture, recently many groundbreaking models for text representation have been proposed. BERT stands for Bidirectional Encoder Representations from Transformers, it was introduced by Devlin et al. (2018) from Google to pre-train deep bidirectional representations from unlabelled text by jointly conditioning on both left and right context in all layers. In the BERT model, several Transformer encoders are stacked on top of each other. Recall the Transformer architecture, the encoder takes an input sequence and the decoder outputs the predicted sequence word by word, since the goal here with BERT is to generate features for downstream NLP tasks, thus the decoder part of the Transformer is discarded. The Transformer model in BERT uses bidirectional self-attention that can read the input sequence from both left-to-right and right-to-left of a word. Such bidirectionality offers an understanding of word relations based on the whole context which makes it one of the core competencies of BERT compare to other language representation models.

BERT is pre-trained on the BooksCorpus (800M words) and the English Wikipedia (2,500M words) using two unsupervised tasks simultaneously shown in Figure 4 a: Masked Language Model (MLM) and Next Sentence Prediction (NSP). In MLM, 15% of the words in the input sequence are randomly replaced by a [MASK] token for BERT to predict given the context while the NSP performs a binary classification to predict whether or not two sentences follow each other given a [SEP] token as a separator between sentences. BERT uses WordPiece (Wu et al. 2016) for tokenization, it has the coverage of 30,000 most frequent and common combinations of words in the vocabulary and any OOV words are broken down into subwords with “##” symbol. For instance, the word ”embedding” would be tokenized as “em”, “##bed”, “##ding”. Two versions of the pre-trained BERT model are made publicly available ³, the base version has 12 encoder layers, 12 attention heads, and 110M parameters whereas the large version has 24 encoder layers, 16 attention heads, and 340M parameters.

[INSERT Figure 4 ABOUT HERE]

³<https://github.com/google-research/bert>

The pre-trained contextual word representations and parameters can be applied to downstream tasks after they have been fine-tuned on task-specific labelled data (Figure 4 b). Fine-tuning is a common step where a classification layer is added to the pre-trained model with all parameters jointly updated, it takes less time and data to develop compared to the pre-training process. The reason behind it is to utilise the pre-trained model to recognise classes they were not originally trained on. Figure 4 (b) shows an example of the fine-tuning procedure for classification tasks such as sentiment analysis. Words from an input sentence are tokenized and passed into the BERT model to produce word embeddings, the special [CLS] token in front of the input sentence treated as a pooled representation of the input is then fed to an output layer for classification. During this process, all parameters are fine-tuned. The architectures of pre-training and fine-tuning are the same except for the output layer, from which the class label probabilities are computed with a standard softmax. With minimal adjustments on the pre-trained model⁴, BERT has obtained new state-of-the-art accuracy on various NLP tasks including sentiment analysis.

6 Experiment

Our empirical study aims to test which of the above Hybrid approaches (lexicon-based, machine learning or BERT) performs best at identifying the sentiment of Glassdoor employee reviews across seven experiments.

Data

Our data comes from *Glassdoor.com*, one of the most extensive job and recruiting platforms. Since its launch in 2008, Glassdoor has collected more than 70 million anonymous reviews by employees. Glassdoor offers reviews in different categories: companies, salaries, and also interviews. In this paper, we focus on the core company reviews. Figure

⁴The Transformer models only require a fine-tuning step on the pre-trained model with a classification layer instead of training a whole separate classifier when performing sentiment analysis. The word embeddings we discussed previously, on the hand, only create feature representation of the text hence they need the help of a separate classifier to deliver the task.

5 demonstrates an anonymous company review from Glassdoor. On starting a review, the user is first asked to identify the company from a drop-down list. Then they are asked to “rate your experience of this company” between one star and five stars. No further guidance is provided on the interpretation of the task or the scale. We take this overall star rating as a proxy for the reviewer’s sentiment towards the company. Referring to the classification above, it is a conceptual and self-reported sentiment label. Reviewers also have to provide text comments with a minimum of five words on *pros* and *cons* for the company. There is an optional text field to provide *advice to management*. To maximise the text available for sentiment analysis, our sample only includes reviews that provide advice to management. The density histogram and word cloud for these three columns are shown in Appendix 2. The majority of the reviews are in short phrases or incomplete sentences. There are additional fields, including six sub-ratings from one star to five stars for *Overall*, *Work-life Balance*, *Culture and Values*, *Career Opportunities*, *Compensation and Benefits*, *Diversity and Inclusion*, and *Senior Leadership*, and “thumbs up or down” questions on the rating of the CEO, whether the reviewer would recommend the company to a friend, and their view of the business outlook for the next six months. All of these are optional and we do not include them in our analysis.

[INSERT Figure 5 ABOUT HERE]

In our analysis, we web-crawled 20,000 employee reviews from S&P 500 companies. Based on the majority of sentiment analysis studies in the NLP field, we believe our sample size has a balanced trade-off between the baseline representation and computational cost. The textual data is cleaned and pre-processed using python’s NLTK package to reduce noise for sentiment analysis. We start with tokenisation where the entire text review is split into individual words. Then we remove stopwords such as “a”, “and”, “or”, and “the”, which occur frequently in the reviews but are not useful for the analysis. Punctuation is also removed, this step is followed by stemming which is removing the suffix from a word and reducing it to its root form. Lastly, we perform parts of speech (POS) tagging to label words with grammatical descriptions, such as nouns, adjectives and verbs etc. such that words are incorporated with context.

Experimental Setup

To perform sentiment analysis, the lexicon-based approaches count the number of positive and negative words in the text and assign the corresponding sentiment, whereas the machine learning and pre-trained models function in a supervised manner and require labelled data during the training process. Many sentiment analyses using review-based social media data, such as Yelp restaurant and hotel reviews, Amazon product reviews, and IMDB movie reviews commonly label the sentiment of text by the overall rating. We use both the text and star ratings from Glassdoor reviews and develop a series of seven experiments to test the performance of different sentiment analysis methods across a variety of tasks. The experiments will identify the best-performing sentiment analysis approach overall but also allow us to see how the performance of the different methods changes as the ambiguity and complexity of the tasks increase. The aim is to inform the choice of method by future researchers depending on their research aims and resources. Table 1 shows a summary of our experiments including the text used in an experiment, and how it is labelled. Positive sentiment is indicated by +, and – represents negative sentiment.

Experiment 1 takes advantage of the fact that Glassdoor employee reviews separate positive and negative comments into *pros* and *cons* fields. The first and simplest task is therefore to see if the sentiment analysis method can correctly identify the text from the *pros* field as representing positive sentiment and the text from the *cons* field as representing negative sentiment. This provides a baseline test of performance in a simple binary sentiment classification. Note that all the reviews contain text for *pros* and *cons* as these were compulsory fields.

The following experiments (2-7) take the overall star rating for the review as the intended sentiment (as the self-reported and subjective assessment of the reviewer) and test how accurately the different methods can predict this sentiment based on the text comments. The experiments differ in terms of the text used and how the sentiment is labelled. Experiment 2 only uses the text from the *pros* and *cons* fields while Experiments 3-7 add the text from *advice to management*. While including *advice to management* will

increase the size of the text corpus available for analysis and so, we might assume, its accuracy, it is also possible that “advice” might be less straightforward and more difficult to interpret than *pros* and *cons*. Since our data collection was limited to those reviews that did complete the optional *advice to management* field, this also includes the full sample.

Experiments 3-7 are designed to test how well the different sentiment analysis methods perform at varying levels of ambiguity and complexity. Experiments 2 and 3 are limited to reviews with an overall rating of one star or five stars - which are likely to represent the strongest expressions of sentiment - and a binary classification as either positive or negative. We expect the sentiment analysis methods to be more accurate when dealing with these cases than in the more ambiguous situations in the subsequent experiments. Note that these experiments have a smaller sample size because of the restriction to one- and five-star reviews as shown in Figure 6.

[INSERT Figure 6 ABOUT HERE]

Experiment 4 widens the definition of positive and negative sentiment by including one and two-star reviews as representing negative sentiment and four- and five-star reviews as representing positive sentiment. We expect that the inclusion of two- and four-star reviews will tend to increase the ambiguity of the sentiments expressed and therefore reduce the performance of the sentiment analysis methods. As a result, the sample size is larger than Experiments 2 and 3 but smaller than Experiments 1 and 5-7.

Finally, Experiments 5-7 incorporate three-star reviews, which are likely to be the most difficult to classify in terms of sentiment. Experiments 5 and 6 retain a binary sentiment classification. In Experiment 5, three-star reviews are classified as positive sentiment (so one and two stars = negative and three, four and five stars = positive). In Experiment 6, three-star reviews are classified as negative sentiment (so one, two and three stars = negative and four and five stars = positive). Experiment 7 introduces a third category of “neutral” for three-star reviews (so one and two stars = negative; three stars = neutral; four and five stars = positive). Our expectation is that the performance

of the models will decline as the degree of ambiguity and complexity increases with the introduction of a multi-class rather than binary sentiment classification.

[INSERT Table 1 ABOUT HERE]

Figure 6 illustrates the class distribution of the experiments, the portion of the positive class is mostly larger than the negative ones. To handle the class imbalance issue, our data is split in a stratified fashion such that the proportions of each class are approximately the same in the training and testing sets. Under each one of the seven experimental settings, we are also interested in the performance of BERT models in a binary and a multi-class sentiment analysis compared to the lexicon-based approaches as well as the machine learning techniques combined with BOW, TF-IDF, Word2Vec and GloVe word embeddings. The hyper-parameters in machine learning methods are optimised through grid search. The description of these models is listed in Appendix 3. In addition to the original cased and uncased BERT models, we also evaluate the performance of FinBERT (Yang et al. 2020, Huang et al. 2022). During fine-tuning of the pre-trained models, we use AdamW optimiser and select the learning rate of $2e-5$, batch size of 32 and training epoch of five. Figure 7 demonstrates the pipeline of the sentiment analysis.

[INSERT Figure 7 ABOUT HERE]

Evaluation Metrics

There are many evaluation metrics for measuring classification performance. We chose the Matthews correlation coefficient (MCC) as the main evaluation metric for the sentiment analysis methods comparison. A primary reason for the choice of MCC is because it excels in dealing with imbalanced class issues in the classification. As indicated in Figure 6, the proportion of sentiment class in Experiments 1, 2, 3 and 6 are relatively close, however, the class distribution is more imbalanced in Experiments 4, 5 and 7 where the differences are more than 20%. We also report other classic evaluation metrics as a robustness check: Precision, Recall, F-1 score and Accuracy.

Given the performance measurements of a confusion matrix: true positive (TP), false negative (FN), true negative (TN), and false positive (FP), these evaluation metrics are interpreted and computed as follows in a binary sentiment classification.

Matthews correlation coefficient (MCC): MCC was introduced by Matthews (1975) to measure the quality of machine learning classifications. In statistics, it is referred to as the phi coefficient or mean square contingency coefficient. It is designed to assess all four of the confusion matrix matrices: TP, FN, TN, and FP regardless of the size of each class, therefore, it handles imbalanced data very well.

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(FN + TN)(FP + TN)(TP + FN)}} \quad (10)$$

Accuracy: It is the most intuitive measure by looking at the overall performance of a model. It tells a ratio of correctly predicted observations to the total observations. However, it works best if the classes are balanced. For example, if the data contains 10% of negative instances and a classifier always assigns the positive label, the overall accuracy would still reach 90% since it would correctly predict 90% instances.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (11)$$

Precision: It is known as the positive predictive value. Precision calculates the TP cases out of the total predicted positives, and it is a good measure for detecting FP.

$$Precision = \frac{TP}{TP + FP} \quad (12)$$

Recall: Unlike precision that ignores all but positive instances and predictions, recall tells the coverage of the actual positive sample. It is also called sensitivity or true positive rate, which indicates the ratio of positive instances that are correctly

detected by the classifier.

$$Recall = \frac{TP}{TP + FN} \quad (13)$$

F-1 Score: It is a weighted average of Precision and Recall. It is a good measure to use when seeking a balance between Precision and Recall, and the value goes up only when both Precision and Recall are high. F-1 Score is usually preferred to accuracy if there is a large uneven class distribution.

$$F1 - score = 2 \times \frac{Precision \times Recall}{Precision + Recall} = \frac{2TP}{2TP + FP + FN} \quad (14)$$

Although F-1 Score can handle class imbalance issues to an extent and is arguably easier to interpret than MCC, Boughorbel et al. (2017), Yao and Shepperd (2020), Chicco and Jurman (2020) and Chicco et al. (2021) prove F-1 score to be more biased and misleading than MCC because it ignores the count of TN and highly influenced by positive observations. In contrast, MCC generates a more balanced assessment by taking into account all four items in the confusion matrix. Therefore, in a case where the majority class is negative, for instance, Experiment 6 in Figure 6, MCC is a fairer option.

The range of MCC is between -1 and 1, a high score is achievable only if the prediction performs well on the majority of data instances from every class, independently of the class ratios in the overall sample. The value of Accuracy, Precision, Recall, and F-1 score ranges from 0 to 1, with a higher score indicating better performance. In a multi-class scenario, these metrics can be computed by one-vs-all comparisons through macro averaging which turns multi-class predictions into multiple sets of binary predictions and then averages the corresponding metric for each of the binary cases.

Result Discussion

The performance of the different sentiment analysis methods in the seven experiments are shown in Tables 2. Across the experiments, the average MCC of lexicon-based, machine

learning and BERT-related approaches are 0.224, 0.507 and 0.674, respectively. According to the mean MCC, we observe that the performance of machine learning approaches was double and BERT-related approaches was triple that of the lexicon-based approaches. In addition, the MCC of BERT-related approaches was also 50% higher than machine learning ones.

[INSERT Table 2 ABOUT HERE]

We discuss the results in more detail below. However, to provide an accessible overview of the performance of the different sentiment analysis methods, Figure 8 presents them in rank order based on our preferred MCC metric. Models on the far right have the highest MCC score on average and hence are ranked the highest.

[INSERT Figure 8 ABOUT HERE]

The first aim of this paper was to identify the best-performing method for sentiment analysis. It is clear that the BERT models were the best performing across all experiments, regardless of the association with various star ratings or whether we used binary and three-class sentiment analysis. As shown in Figure 8, looking at the maximum mean MCCs from the three types of sentiment classifiers, BERT models had the highest mean MCC of 0.91. The best performing machine learning approach had the highest mean MCC of 0.77. There is considerable variation in the performance of different combinations of word embeddings and classifiers, and across the different experiments. However, all the machine learning approaches performed worse than the BERT and better than the lexicon-based approaches (highest mean MCC = 0.55), which are ranked the lowest on average.

To consider the BERT models in more detail, normally the choice of BERT-base-case or BERT-base-uncased is dependent on whether the letter case is sensitive to the task. From our observation of the raw text, the reviews are usually short phrases and incomplete sentences in lowercase. As a result, in Experiments 1 and 2, uncased BERT has a slight advantage over cased. FinBERT is a BERT model pre-trained on financial communication text, it works very well on employee reviews containing financial terms.

After a closer examination of the BERT models' misclassified cases, we find the comment "None that I can think of" or sarcastic expressions frequently appeared in *pros* or *cons*. It is difficult to assign the sentiment to these cases without human intervention, but we noticed that FinBERT handles these cases slightly better than BERT. This may say something about the level of sarcasm or cynicism in financial language. Nevertheless, we find after including the additional text from *advice to management* in Experiments 3 to 6, the performance of FinBERT declines noticeably, probably due to the fact that the *advice to management* contains fewer finance-related words. For this reason, BERT should be preferred when using non-financial texts.

Turning to Machine Learning approaches, Figure 9 reports the average of MCC grouped by word embeddings and machine learning classifiers. This aggregated matrix identifies that while RF can be used as a baseline method, NB and DT have less predicting power compared to LR, SVM and XGB. Word2Vec and GloVe did not show a clear advantage over the simple feature representations such as BOW and TF-IDF even though theoretically the task should benefit from the semantic meaning of word embeddings such as Word2Vec and GloVe. Mishev et al. (2020) have also found that TF-IDF is the best text representation method among BOW, Word2Vec and GloVe testing on the Financial Phrase Bank (Malo et al. 2014) and SemEval2017-Task5 dataset (Cortis et al. 2017). A possible explanation is that pre-trained word embeddings are usually high-dimensional, and machine learning methods struggle to model all of the information included.

[INSERT Figure 9 ABOUT HERE]

Despite their relatively poor performance in our experiments, lexicon-based approaches may still have value, particularly when the speed of computing is an issue or when dealing with unlabelled text. VADER was the best-performing lexicon-based approach, the LM dictionary designed for financial texts performed better than the HIV4 dictionary and SentiWordNet, which matches the results from previous studies. Before we performing the grid search to optimise hyper-parameters in machine learning approaches, we find VADER outperformed all DT in Experiments 2, 5 and 6 (results of DT with untuned

hyper-parameters are displayed in Appendix 4). This suggests that, although the dictionary approaches are weak, they have the potential to improve if the dictionaries can be tuned subject to the task as well.

In addition to the comparison of sentiment analysis methods, we were also interested in the impact of ambiguity of labelling and complexity of classification on performance. Therefore, we aggregate the performance of the same type of model. Figure 10 displays the change of mean MCC when vary how the star ratings are used to provide sentiment labels.

[INSERT Figure 10 ABOUT HERE]

Experiment 1 serves as a baseline case. In Experiment 1, sentiment was identified based on the text from the *pros* and *cons* fields separately. In effect, could the sentiment analysis method correctly identify the text from *pros* as expressing positive sentiment and that from *cons* negative? This was the most straightforward task and all methods performed their best. The BERT models have a mean MCC of 0.91 in predicting that text from the *pros* field represented positive sentiment and text from the *cons* field negative sentiment. As noted above, some sarcastic responses may account for some of the errors. The machine learning method and the lexicon-based approach both achieved their highest MCC of 0.70 and 0.44, respectively.

The next set of experiments were more complex. In Experiment 2, we tested whether the methods could identify one-star reviews as negative and five-star reviews as positive based on the combined text from the *pros* and *cons* fields. We observed a relative decrease of 9% - 34% in mean MCC for all types of approaches from Experiment 1 to Experiment 2. The reduced mean MCC reveals that using the overall rating as the intended sentiment can potentially introduce more noise to the models ⁵. Experiment 3 included the text from *advice to management*. As the amount of text increases, the BERT models and the machine learning models benefit from the additional training data and therefore, their

⁵It is unlikely that the main factor in the decrease in mean MCC in Experiments 2 and 3 compared to Experiment 1 was due to the reduced sample size after restraining the star ratings, because the mean MCC continued to decrease when the sample size was increased in the following experiments.

mean MCC increased relatively by 1% and 9%, respectively. By contrast, longer texts on average reduce the MCC of the lexicon-based approach by relatively 3%.

As was pointed out in the experiment setups, we were curious about the impact of increasing the ambiguity of the sentiment by including first two and four star ratings, and then three star overall ratings. Experiments 4 to 7 used the texts from *pros*, *cons* and *advice to management* but changed how the star ratings were used to label the review sentiment as positive or negative. In Experiment 4, we included one and two star ratings as negative and four and five star ratings as positive. Adding two and four star ratings should have included more ambiguous and difficult to classify cases. Indeed, we see a sizeable drop in mean MCC for all types of methods, a drop that continues after we introduced the three-star ratings - the most ambiguous category in terms of sentiment. From Experiments 3 to 5, the mean MCC dropped relatively around 40% for all types of methods. The first point is that, when more ambiguous labels were included, performance dropped. Second, ratings in the middle ground are more ambiguous representations of textual sentiment than ratings at the extremes (one or five stars).

Experiments 5 and 6 treated three-star ratings as positive and negative respectively. The models had slightly higher MCC on average when three-star reviews were considered as being positive as compared to negative. One interpretation could be that the underlying sentiment of three-star ratings could be seen as grudgingly positive or at least not as negative. The other explanation is that there is discrepancy between a review's sentiment as indicated by its overall star rating and by the contents of text comments. Experiment 7 is the most complex of all, introducing a third sentiment class (neutral) as well as positive and negative. The mean MCC of BERT models drops from 0.528 in Experiment 5 to 0.436 in Experiment 7, the machine learning models from 0.384 to 0.326, and lexicon-based approaches from 0.173 to only 0.024. Given this decline in performance across the board, researchers should be cautious when applying any sentiment analysis method to more ambiguously-labelled data or to produce multi-class sentiment classifications.

We have looked at the accuracy of models at predicting the user-assigned sentiment

from their text comments but this ignores the possibility that there is additional information that could be extracted by looking at the contents of the text fields themselves. The preceding analysis has assumed that the user-generated star rating is the best guide to the intended sentiment of the review, in order to test the accuracy of different sentiment analysis methods at predicting this sentiment from the associated text comments. However, it is possible that, in some cases, the text comments could contain valuable additional information that could be extracted. Therefore, solely focusing on numerical ratings as measures of sentiment risks missing significant information that could be gleaned from text, potentially providing more insight and context to deepen the research findings.

7 Conclusion

In summary, this paper offers a comprehensive assessment of 31 frequently-used sentiment analysis methods used by accounting and finance academics, from the lexicon-based approach, to machine learning classifiers with word embeddings, and then to the powerful NLP model BERT. These different approaches were classified within a framework that consists of the Pure Conceptual Approach, the Hybrid Approach, and the Pure Empirical Approach for detecting the sentiment of textual data. The paper focuses on the Hybrid approach, which is divided into methods employing a Deductive Encoding of Meaning (e.g. lexicon-based approaches) and an Inductive Encoding of Meaning (e.g. machine learning and pre-trained models).

Our work appears to be the first study to evaluate these sentiment analysis methods using the text comments within Glassdoor employee reviews. Our empirical study has two main purposes. First, we compare the relative performance of the different sentiment analysis methods to help future researchers to identify the optimal method for their work. The results conclusively prove that BERT models produce the best outcomes in both binary and three-class sentiment classification. However, their deployment requires a higher level of computing power. The machine learning methods produce solid pre-

dictionaries, and their performance is better when the model has better tolerance towards high-dimensional feature representations. We show that LR, SVM, and XBG are reliable classifiers, and the choice of word embedding is as important. Among them, our results indicate that word embeddings such as Word2Vec and GloVe do not show a clear advantage over BOW and TF-IDF due to their high dimensionality. In collaboration with machine learning models, future research could consider exploring their usage along with more advanced classifiers such as RNNs, Convolutional Neural Networks (CNNs), and Long Short Term Memory (LSTM). Lexicon-based approaches are the weakest performers but may still have some use where computing power or expertise is at a premium and when it is not feasible to manually annotate a training sample. In some cases, VADER produces very close results to DT and NB.

Secondly, we demonstrate the performance of all the sentiment analysis models declines as the complexity of the task and ambiguity of the labels increase. All methods perform their best in identifying the text from the *pros* field as representing positive sentiment and the text from the *cons* field as representing negative sentiment. However, when we take the overall star rating of the reviews as the expected sentiment, the performance of all methods decreases. This is especially the case for the three-star ratings, which contain the most ambiguous expressions of sentiment. Finally, we show that the complexity of the sentiment classification increases with the introduction of neutral sentiment and therefore, the performance of the model decreases.

Analysing Glassdoor employee reviews with advanced NLP methods is important in accounting and finance because it allows organisations to gain valuable insights into the sentiment, concerns, and feedback of their employees. This information can be used to improve workplace culture, address any issues, and retain top talent. Additionally, analysing employee reviews can provide insight into how the organisation is perceived by the market, which can impact its reputation and ability to attract new clients and investors. By using advanced NLP methods, organisations can more accurately and effectively analyse employee reviews and take action based on the insights obtained.

Our experiments also present an opportunity for future studies to dive deeper into

Glassdoor employee reviews. Future research could examine whether the numerical rating of reviews fully convey the sentiment in their textual contents. In addition, our comparison of different models can provide a reference for future research on sentiment analysis of financial texts. The state-of-the-art NLP models are more powerful but are more computationally expensive at the same time. Depending on the specific task, researchers can choose a model suitable for them considering the complexity and accuracy trade-off.

References

- Abbasi, A., J. Li, D. Adjero, M. Abate, and W. Zheng. 2019. Don't mention it? Analyzing user-generated content signals for early adverse event warnings. *Information Systems Research* 30(3), 1007–1028.
- Antweiler, W. and M. Z. Frank. 2004. Is all that talk just noise? The information content of internet stock message boards. *The Journal of Finance* 59(3), 1259–1294.
- Araci, D.. 2019. FinBERT: Financial sentiment analysis with pre-trained language models. *arXiv preprint arXiv:1908.10063*.
- Au, S., M. Dong, and A. Tremblay. 2021. Employee flexibility, exogenous risk, and firm value. *Journal of Financial and Quantitative Analysis* 56(3), 853–884.
- Baccianella, S., A. Esuli, and F. Sebastiani. 2010. Sentiwordnet 3.0: An enhanced lexical resource for sentiment analysis and opinion mining. In *Proceedings of the Seventh International Conference on Language Resources and Evaluation (LREC'10)*.
- Bingler, J. A., M. Kraus, M. Leippold, and N. Webersinke. 2022. Cheap talk and cherry-picking: What ClimateBert has to say on corporate climate risk disclosures. *Finance Research Letters* 47, 102776.
- Bochkay, K., S. V. Brown, A. J. Leone, and J. W. Tucker. 2022. Textual analysis in accounting: What's next? *Contemporary Accounting Research* forthcoming.
- Boughorbel, S., F. Jarray, and M. El-Anbari. 2017. Optimal classifier for imbalanced data using matthews correlation coefficient metric. *PloS One* 12(6), e0177678.
- Breiman, L.. 2001. Random forests. *Machine Learning* 45(1), 5–32.
- Campbell, D. W. and R. Shang. 2022. Tone at the bottom: Measuring corporate misconduct risk from the text of employee reviews. *Management Science* 68(9), 7034–7053.
- Canning, E. A., M. C. Murphy, K. T. Emerson, J. A. Chatman, C. S. Dweck, and L. J. Kray. 2020. Cultures of genius at work: Organizational mindsets predict cultural norms, trust, and commitment. *Personality and Social Psychology Bulletin* 46(4), 626–642.
- Chen, T. and C. Guestrin. 2016. Xgboost: A scalable tree boosting system. In *Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining*, pp. 785–794.
- Chicco, D. and G. Jurman. 2020. The advantages of the Matthews correlation coefficient (MCC) over F1 score and accuracy in binary classification evaluation. *BMC Genomics* 21(1), 6.
- Chicco, D., N. Tötsch, and G. Jurman. 2021. The matthews correlation coefficient (mcc) is more reliable than balanced accuracy, bookmaker informedness, and markedness in two-class confusion matrix evaluation. *BioData Mining* 14(1), 1–22.
- Corritore, M., A. Goldberg, and S. B. Srivastava. 2020. Duality in diversity: How intrapersonal and interpersonal cultural heterogeneity relate to firm performance. *Administrative Science Quarterly* 65(2), 359–394.
- Cortes, C. and V. Vapnik. 1995. Support-vector networks. *Machine Learning* 20(3), 273–297.
- Cortis, K., A. Freitas, T. Daudert, M. Huerlimann, M. Zarrouk, S. Handschuh, and B. Davis. 2017, August. SemEval-2017 task 5: Fine-grained sentiment analysis on financial microblogs and news. In *Proceedings of the 11th International Workshop on Semantic Evaluation (SemEval-2017)*, Vancouver, Canada, pp. 519–535. Association for Computational Linguistics.
- Creek, S. A., K. M. Kuhn, and A. Sahaym. 2019. Board diversity and employee satisfaction: The mediating role of progressive programs. *Group & Organization Management* 44(3), 521–548.

- Dabirian, A., J. Kietzmann, and H. Diba. 2017. A great place to work!? Understanding crowdsourced employer branding. *Business Horizons* 60(2), 197–205.
- Dabirian, A., J. Paschen, and J. Kietzmann. 2019. Employer branding: Understanding employer attractiveness of it companies. *IT Professional* 21(1), 82–89.
- Devlin, J., M.-W. Chang, K. Lee, and K. Toutanova. 2018. BERT: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*.
- Engelberg, J. E., A. V. Reed, and M. C. Ringgenberg. 2012. How are shorts informed?: Short sellers, news, and information processing. *Journal of Financial Economics* 105(2), 260–278.
- Esuli, A. and F. Sebastiani. 2006. Sentiwordnet: A publicly available lexical resource for opinion mining. In *Proceedings of the Fifth International Conference on Language Resources and Evaluation (LREC'06)*.
- Feldman, R., S. Govindaraj, J. Livnat, and B. Segal. 2010. Management's tone change, post earnings announcement drift and accruals. *Review of Accounting Studies* 15(4), 915–953.
- Feuerriegel, S. and J. Gordon. 2019. News-based forecasts of macroeconomic indicators: A semantic path model for interpretable predictions. *European Journal of Operational Research* 272(1), 162–175.
- Frankel, R., J. Jennings, and J. Lee. 2022. Disclosure Sentiment: Machine Learning vs. Dictionary Methods. *Management Science* 68(7), 5514–5532.
- Godbole, N., M. Srinivasaiah, and S. Skiena. 2007. Large-scale sentiment analysis for news and blogs. *The International AAAI Conference on Web and Social Media (ICWSM)* 7(21), 219–222.
- González-Carvajal, S. and E. C. Garrido-Merchán. 2020. Comparing BERT against traditional machine learning text classification. *arXiv preprint arXiv:2005.13012*.
- Hagenau, M., M. Liebmann, and D. Neumann. 2013. Automated news reading: Stock price prediction based on financial news using context-capturing features. *Decision Support Systems* 55(3), 685–697.
- Hales, J., J. R. Moon Jr, and L. A. Swenson. 2018. A new era of voluntary disclosure? empirical evidence on how employee postings on social media relate to future corporate disclosures. *Accounting, Organizations and Society* 68, 88–108.
- Hickman, L., S. Thapa, L. Tay, M. Cao, and P. Srinivasan. 2022. Text preprocessing for text mining in organizational research: Review and recommendations. *Organizational Research Methods* 25(1), 114–146.
- Huang, A. H., H. Wang, and Y. Yang. 2022. FinBERT: A large language model for extracting information from financial text. *Contemporary Accounting Research* forthcoming.
- Huang, A. H., A. Y. Zang, and R. Zheng. 2014. Evidence on the information content of text in analyst reports. *The Accounting Review* 89(6), 2151–2180.
- Huang, K., M. Li, and S. Markov. 2020. What do employees know? Evidence from a social media platform. *The Accounting Review* 95(2), 199–226.
- Huang, M., P. Li, F. Meschke, and J. P. Guthrie. 2015. Family firms, employee satisfaction, and corporate performance. *Journal of Corporate Finance* 34, 108–127.
- Hutto, C. and E. Gilbert. 2014. VADER: A parsimonious rule-based model for sentiment analysis of social media text. In *Proceedings of the International AAAI Conference on Web and Social Media*, Volume 8, pp. 216–225.
- Jaggi, M., P. Mandal, S. Narang, U. Naseem, and M. Khushi. 2021. Text mining of stocktwits data for predicting stock prices. *Applied System Innovation* 4(1), 13.
- Jing, C., K. Keasey, I. Lim, and B. Xu. 2019. Financial constraints and employee satisfaction. *Economics Letters* 183, 108599.

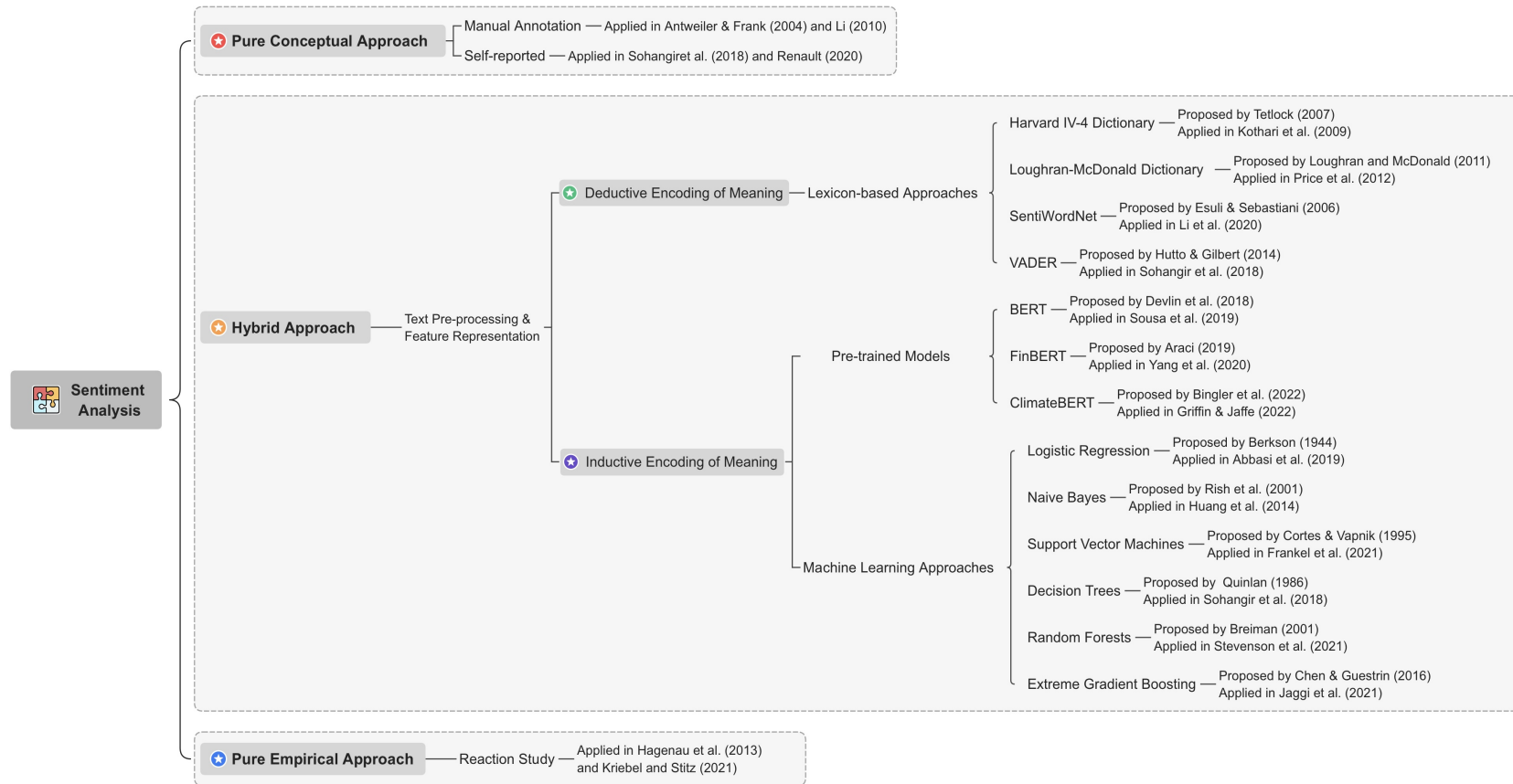
- Kobayashi, V. B., S. T. Mol, H. A. Berkers, G. Kismihok, and D. N. Den Hartog. 2018. Text classification for organizational researchers: A tutorial. *Organizational Research Methods* 21(3), 766–799.
- Kothari, S. P., X. Li, and J. E. Short. 2009. The effect of disclosures by management, analysts, and business press on cost of capital, return volatility, and analyst forecasts: A study using content analysis. *The Accounting Review* 84(5), 1639–1670.
- Kriebel, J. and L. Stitz. 2021. Credit default prediction from user-generated text in peer-to-peer lending using deep learning. *European Journal of Operational Research* 302(1), 309–323.
- Li, F.. 2010. The information content of forward-looking statements in corporate filings—a naïve bayesian machine learning approach. *Journal of Accounting Research* 48(5), 1049–1102.
- Liu, Z., D. Huang, K. Huang, Z. Li, and J. Zhao. 2021. FinBERT: A pre-trained financial language representation model for financial text mining. In *Proceedings of the Twenty-Ninth International Conference on International Joint Conferences on Artificial Intelligence*, pp. 4513–4519.
- Loughran, T. and B. McDonald. 2011. When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks. *The Journal of Finance* 66(1), 35–65.
- Malo, P., A. Sinha, P. Korhonen, J. Wallenius, and P. Takala. 2014. Good debt or bad debt: Detecting semantic orientations in economic texts. *Journal of the Association for Information Science and Technology* 65(4), 782–796.
- Matthews, B. W.. 1975. Comparison of the predicted and observed secondary structure of t4 phage lysozyme. *Biochimica et Biophysica Acta (BBA)-Protein Structure* 405(2), 442–451.
- Melián-González, S., J. Bulchand-Gidumal, and B. González López-Valcárcel. 2015, jan. New evidence of the relationship between employee satisfaction and firm economic performance. *Personnel Review* 44(6), 906–929.
- Mikolov, T., K. Chen, G. Corrado, and J. Dean. 2013. Efficient estimation of word representations in vector space. *arXiv preprint arXiv:1301.3781*.
- Miller, G. A.. 1995. WordNet: A lexical database for english. *Communications of the ACM* 38(11), 39–41.
- Mishev, K., A. Gjorgjevikj, I. Vodenska, L. T. Chitkushev, and D. Trajanov. 2020. Evaluation of sentiment analysis in finance: from lexicons to transformers. *IEEE Access* 8, 131662–131682.
- Nassirtoussi, A. K., S. Aghabozorgi, T. Y. Wah, and D. C. L. Ngo. 2014. Text mining for market prediction: A systematic review. *Expert Systems with Applications* 41(16), 7653–7670.
- Pandey, S. and S. K. Pandey. 2019. Applying natural language processing capabilities in computerized textual analysis to measure organizational culture. *Organizational Research Methods* 22(3), 765–797.
- Pennington, J., R. Socher, and C. D. Manning. 2014. GloVe: Global vectors for word representation. In *Proceedings of the 2014 conference on empirical methods in natural language processing (EMNLP)*, pp. 1532–1543.
- Price, S. M., J. S. Doran, D. R. Peterson, and B. A. Bliss. 2012. Earnings conference calls and stock returns: The incremental informativeness of textual tone. *Journal of Banking & Finance* 36(4), 992–1011.
- Quinlan, J. R.. 1986. Induction of decision trees. *Machine Learning* 1(1), 81–106.
- Renault, T.. 2020. Sentiment analysis and machine learning in finance: a comparison of methods and models on one million messages. *Digital Finance* 2(1), 1–13.
- Rish, I. et al.. 2001. An empirical study of the naive bayes classifier. In *IJCAI 2001 workshop on empirical methods in artificial intelligence*, Volume 3, pp. 41–46.

- Robertson, J., S. Lord Ferguson, T. Eriksson, and A. Näppä. 2019. The brand personality dimensions of business-to-business firms: a content analysis of employer reviews on social media. *Journal of Business-to-business Marketing* 26(2), 109–124.
- Rogers, J. L., A. Van Buskirk, and S. L. Zechman. 2011. Disclosure tone and shareholder litigation. *The Accounting Review* 86(6), 2155–2183.
- Schmiedel, T., O. Müller, and J. vom Brocke. 2019. Topic modeling as a strategy of inquiry in organizational research: A tutorial with an application example on organizational culture. *Organizational Research Methods* 22(4), 941–968.
- Sohangir, S., N. Petty, and D. Wang. 2018. Financial sentiment lexicon analysis. In *2018 IEEE 12th International Conference on Semantic Computing (ICSC)*, pp. 286–289. IEEE.
- Sousa, M. G., K. Sakiyama, L. de Souza Rodrigues, P. H. Moraes, E. R. Fernandes, and E. T. Matsubara. 2019. BERT for stock market sentiment analysis. In *2019 IEEE 31st International Conference on Tools with Artificial Intelligence (ICTAI)*, pp. 1597–1601. IEEE.
- Stamolampros, P., N. Korfiatis, K. Chalvatzis, and D. Buhalis. 2019. Job satisfaction and employee turnover determinants in high contact services: Insights from employees' online reviews. *Tourism Management* 75, 130–147.
- Stevenson, M., C. Mues, and C. Bravo. 2021. The value of text for small business default prediction: A deep learning approach. *European Journal of Operational Research* 295(2), 758–771.
- Stone, P. J. and E. B. Hunt. 1968. The general inquirer: A computer approach to content analysis. *American Journal of Sociology* 73(5), 634–635.
- Storer, A. and A. Reich. 2021. 'Losing My Raise': minimum wage increases, status loss and job satisfaction among low-wage employees. *Socio-Economic Review* 19(2), 681–709.
- Sull, D., C. Sull, and B. Zweig. 2022. Toxic culture is driving the great resignation. *MIT Sloan Management Review* 63(2), 1–9.
- Symitsi, E., P. Stamolampros, and G. Daskalakis. 2018. Employees' online reviews and equity prices. *Economics Letters* 162, 53–55.
- Symitsi, E., P. Stamolampros, G. Daskalakis, and N. Korfiatis. 2021. The informational value of employee online reviews. *European Journal of Operational Research* 288(2), 605–619.
- Tambe, P., X. Ye, and P. Cappelli. 2020. Paying to program? Engineering brand and high-tech wages. *Management Science* 66(7), 3010–3028.
- Tetlock, P. C.. 2007. Giving content to investor sentiment: The role of media in the stock market. *The Journal of Finance* 62(3), 1139–1168.
- Tetlock, P. C., M. Saar-Tsechansky, and S. Macskassy. 2008. More than words: Quantifying language to measure firms' fundamentals. *The Journal of Finance* 63(3), 1437–1467.
- Tsai, M.-F. and C.-J. Wang. 2017. On the risk prediction and analysis of soft information in finance reports. *European Journal of Operational Research* 257(1), 243–250.
- van der Heijden, H.. 2022. Predicting industry sectors from financial statements: An illustration of machine learning in accounting research. *The British Accounting Review* 54(5), 101096.
- Vaswani, A., N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin. 2017. Attention is all you need. *Advances in Neural Information Processing Systems* 30, 5998–6008.
- Wolter, J. S., D. Bock, J. Mackey, P. Xu, and J. S. Smith. 2019. Employee satisfaction trajectories and their effect on customer satisfaction and repatronage intentions. *Journal of the Academy of Marketing Science* 47(5), 815–836.
- Wu, Y., M. Schuster, Z. Chen, Q. V. Le, M. Norouzi, W. Macherey, M. Krikun, Y. Cao, Q. Gao, K. Macherey, et al.. 2016. Google's neural machine translation system: Bridging the gap between human and machine translation. *arXiv preprint arXiv:1609.08144*.

- Xu, Y., M. Armony, and A. Ghose. 2021. The interplay between online reviews and physician demand: An empirical investigation. *Management Science* 67(12), 7344–7361.
- Yang, Y., M. C. S. Uy, and A. Huang. 2020. FinBERT: A pretrained language model for financial communications. *arXiv preprint arXiv:2006.08097*.
- Yao, J. and M. Shepperd. 2020. Assessing software defection prediction performance: Why using the matthews correlation coefficient matters. In *Proceedings of the Evaluation and Assessment in Software Engineering*, EASE '20, New York, NY, USA, pp. 120–129. Association for Computing Machinery.
- Yekini, L. S., T. P. Wisniewski, and Y. Millo. 2016. Market reaction to the positiveness of annual report narratives. *The British Accounting Review* 48(4), 415–430.
- Yukselturk, O. and J. Tucker. 2015. The impact of analyst sentiment on uk stock recommendations and target prices. *Accounting and Business Research* 45(6-7), 869–904.
- Zhao, L., L. Li, X. Zheng, and J. Zhang. 2021. A BERT based sentiment analysis and key entity detection approach for online financial texts. In *2021 IEEE 24th International Conference on Computer Supported Cooperative Work in Design (CSCWD)*, pp. 1233–1238. IEEE.

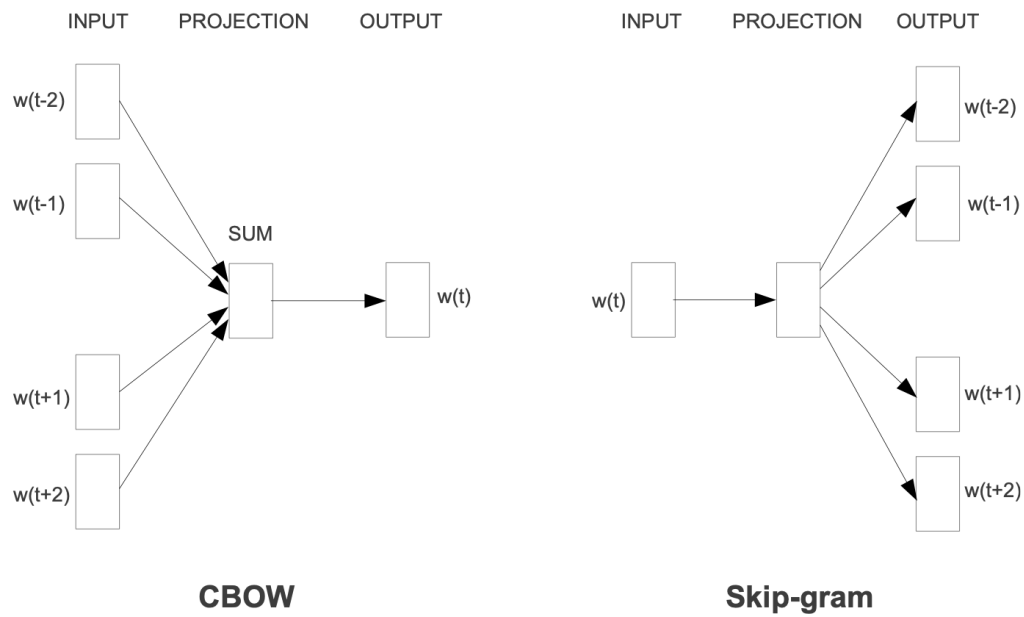
Figures and Tables

Figure 1: Sentiment analysis framework



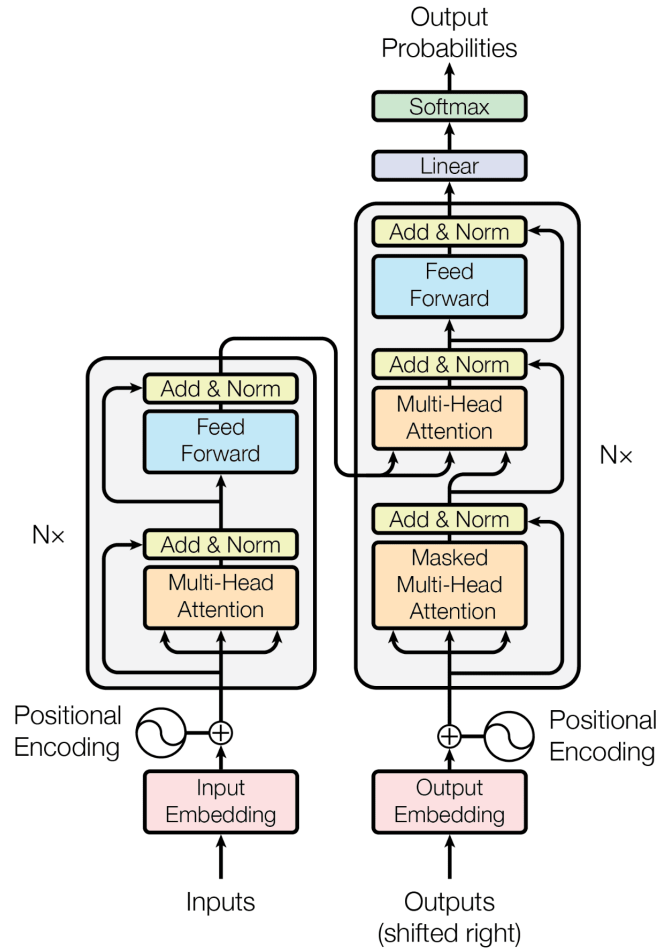
Notes: (1) Pre-trained Models refer to models that have been pre-trained on a large amount of text to gain semantic understandings of words, for instance, the BERT-related models. (2) Machine Learning Approach is a category that encodes features of a sample text that is a part of the corpus.

Figure 2: The Word2Vec architectures



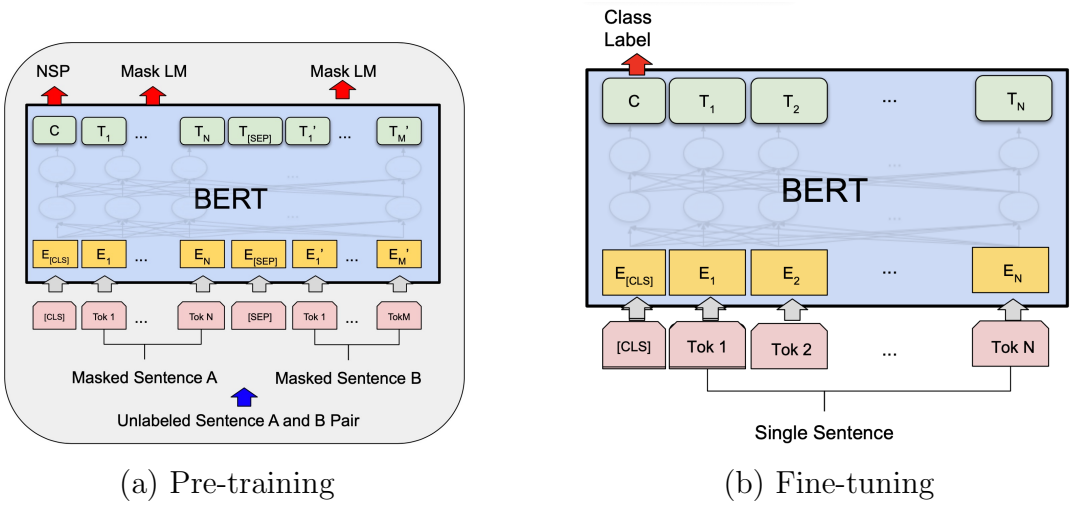
Notes: CBOW: A neural network that learns from the context words to predict the centre word. Skip-Gram: A neural network that learns from the centre word to predict its context words.

Figure 3: The Transformer model architecture



Notes: The left half of this figure is the encoder which builds a continuous word embedding for sequential representation. It is then fed into the decoder (the right half) to generate an output sequence. The output embedding comes from the embedding layer which is shared with the input embedding, and the positional encoding is to make sure the model learns the order of the sequence. The encoder and decoder are composed of a stack of six identical layers, and each layer mainly consists of a feed-forward network and a multi-head attention module.

Figure 4: Overall pre-training and fine-tuning procedures for BERT.



Notes: (a) shows the two tasks for BERT pre-training: Masked Language Model (MLM) and Next Sentence Prediction (NSP). (b) shows how BERT is fine-tuned to a downstream task, this example is for sentiment analysis.

Figure 5: An anonymous employee review from *Glassdoor.com*

4.0 ★★★★★ ✓

Former Employee

Amazing company, but getting big and a bit bureaucratic over time

17 Aug 2017 - Anonymous Employee

✓ Recommend ✗ CEO Approval ✓ Business Outlook

Pros

Amazing culture, perks, growth mindset, most employees are very capable

Cons

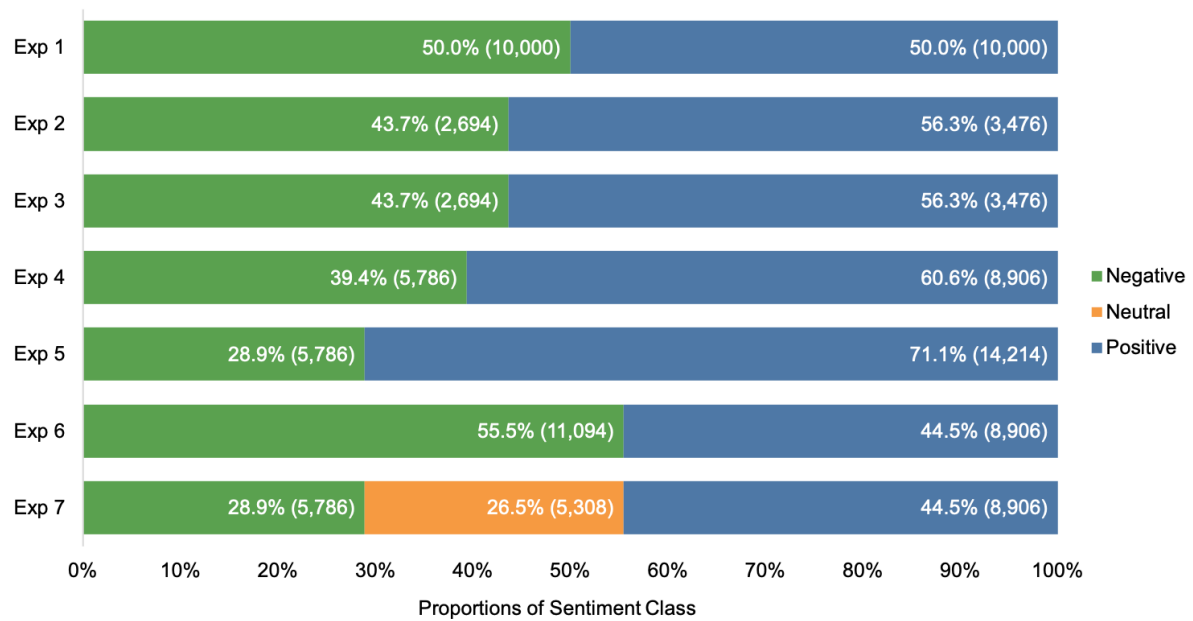
Company has reached a size and a complexity where most roles even at senior level are rather narrow in scope, and there is a lot of "work about work" - Agreeing who does what, defining processes - One can grow impatient and weary

Advice to Management

Streamline the business, simplify the organization, create more autonomous units with clearer accountability

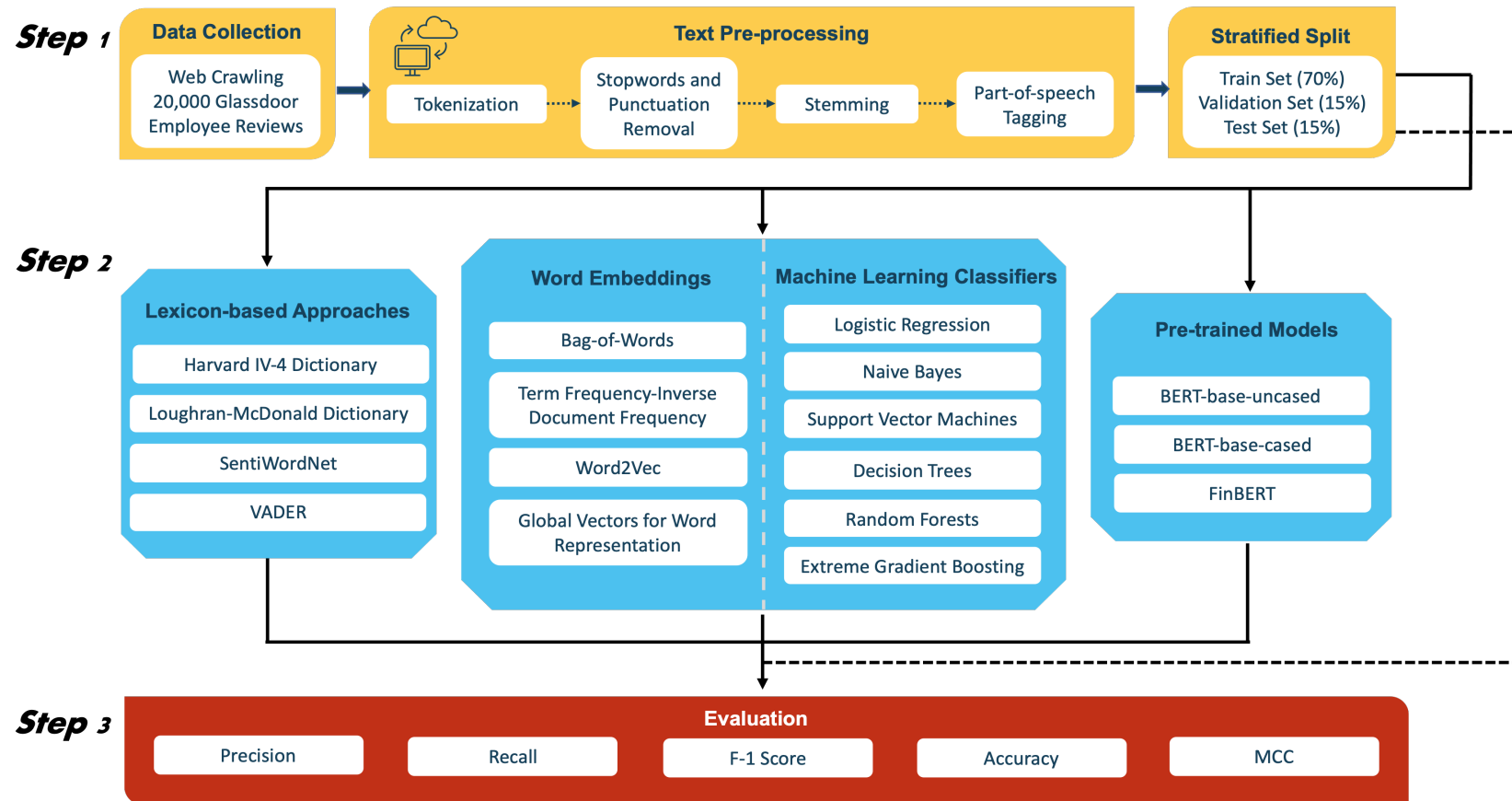
Note: This is a screenshot of a piece of anonymous employee review from Glassdoor. The sub-ratings are optional and not explicitly linked to the overall rating. Reviewers have to leave text comments of at least five words for *pros* and *cons*, and have the option to provide advice to management.

Figure 6: Sentiment class proportions of experiments



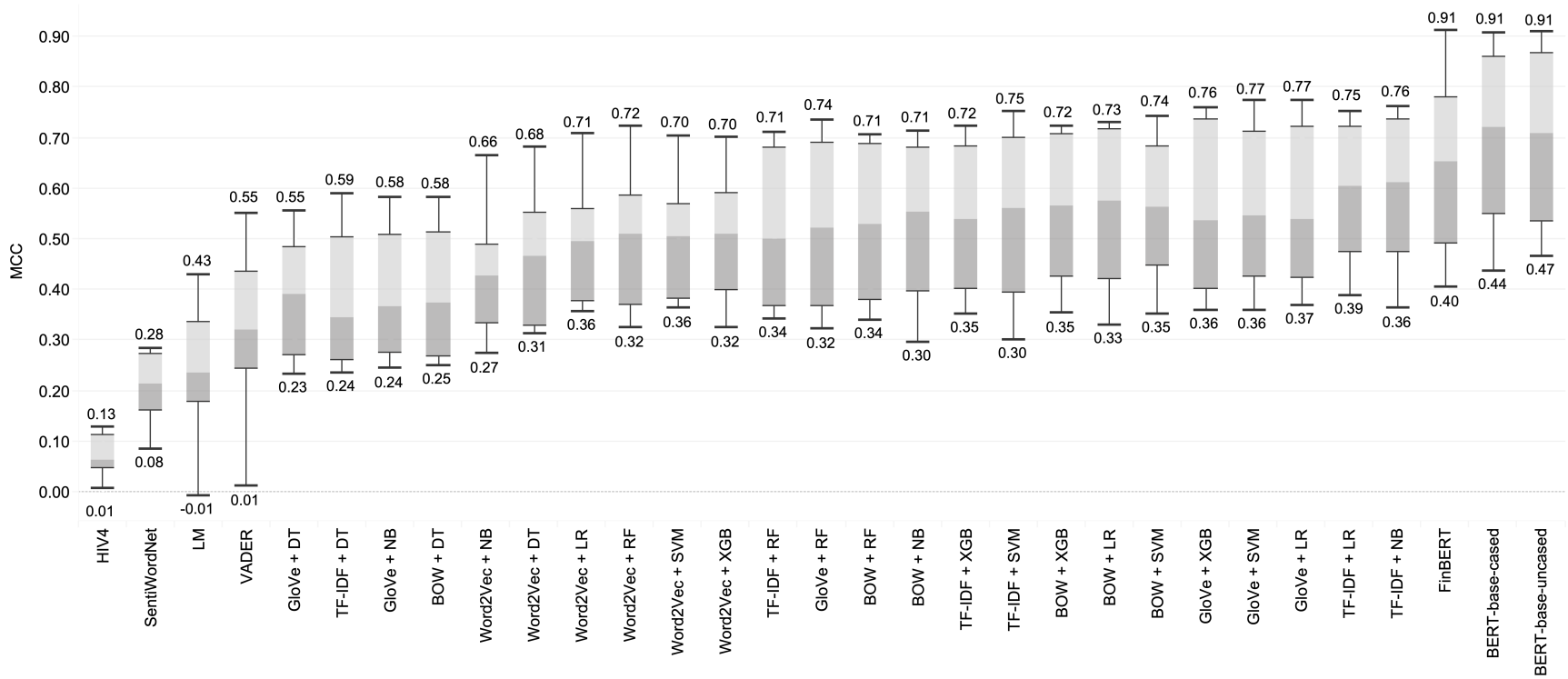
Note: Our sample data consists of 20,000 reviews, This bar chart shows the proportion of sentiment class in each experiment, and the number of observations is included in the parentheses. The proportion of sentiment class in Experiments 1, 2, 3 and 6 are relatively close, however, it is more imbalanced in Experiments 4, 5 and 7 where the differences are more than 20%.

Figure 7: Sentiment analysis experiments pipeline



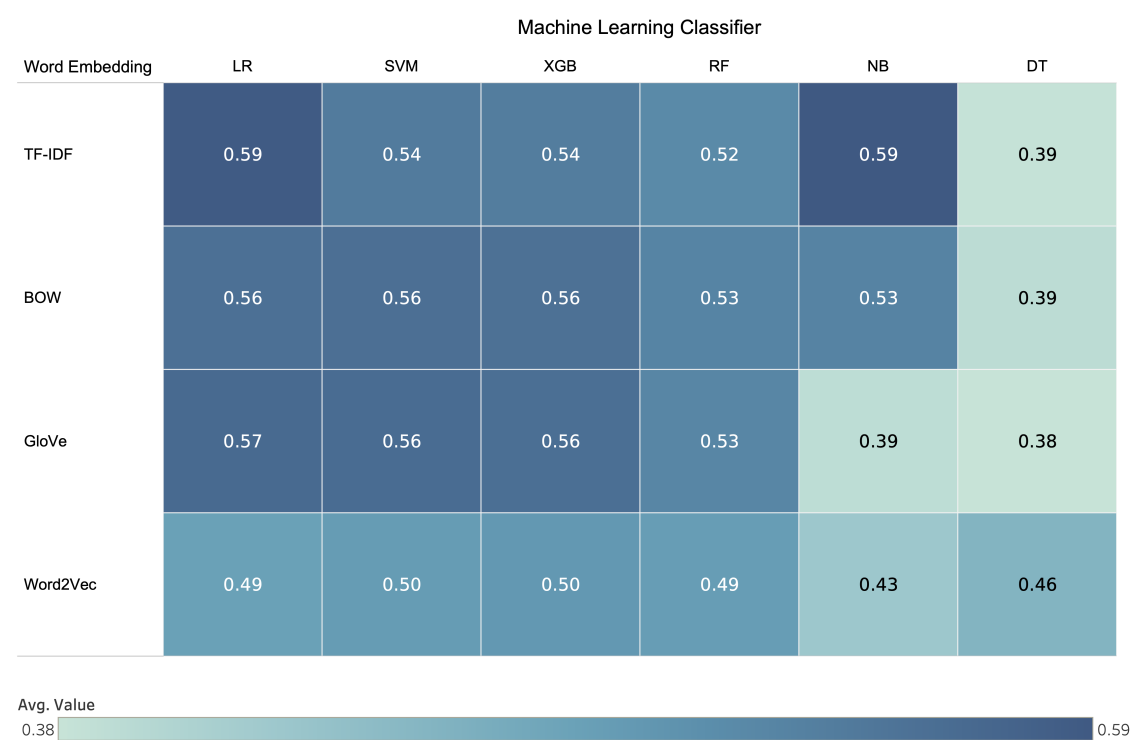
Notes: We first collect the data, pre-process the text and split them into the train, validation and test sets. Next, we fit the data into different sentiment classifiers and eventually evaluate them on the test dataset.

Figure 8: Performance of sentiment analysis methods



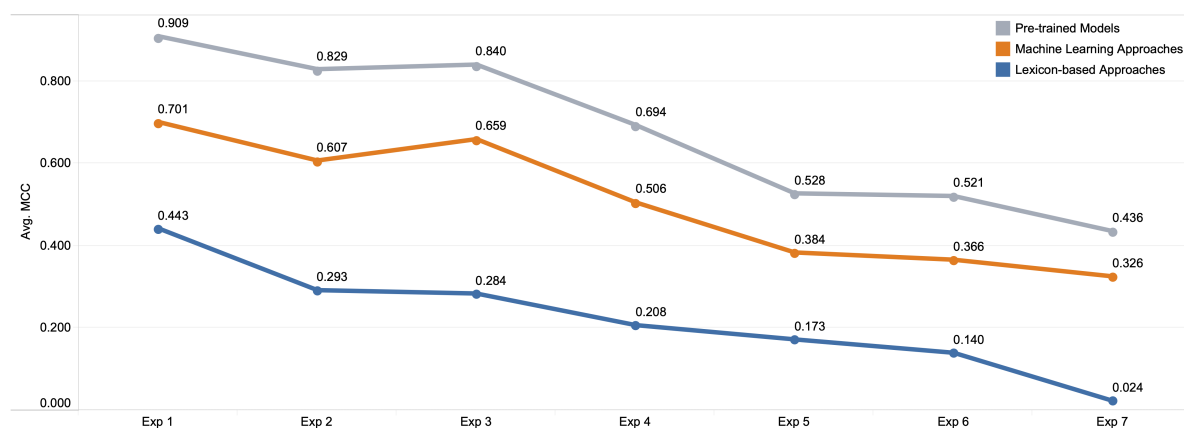
Notes: This box plot shows the performance of sentiment analysis methods in MCC. Each box is composed of the MCC scores of all experiments in which it is placed, the x-axis is sorted by the rank of their mean MCC. The whiskers are extended to values within 1.5 times the interquartile range. The values above and below the whiskers represent the highest and the lowest MCC of a method.

Figure 9: Performance of machine learning classifiers in relation to word embeddings



Notes: This matrix shows the correlation between the machine learning classifiers and word embeddings in terms of their mean MCC.

Figure 10: Change of Avg. MCC per experiment (grouped by the types of sentiment analysis methods)



Notes: This figure shows the change of the mean MCC per experiment after grouping the models from each sentiment analysis method.

Table 1: Experimental setup

	Text			Label				
	Pros	Cons	Advice to Management	Overall Rating: 1 Star	Overall Rating: 2 Star	Overall Rating: 3 Star	Overall Rating: 4 Star	Overall Rating: 5 Star
Exp 1	✓, +	✓, −						
Exp 2	✓	✓		−				+
Exp 3	✓.	✓	✓	−				+
Exp 4	✓	✓	✓	−	−		+	+
Exp 5	✓	✓	✓	−	−	+	+	+
Exp 6	✓	✓	✓	−	−	−	+	+
Exp 7	✓	✓	✓	−	−	Neutral	+	+

Notes: This table summarises the setup of our experiments. The left side of the table suggests the part of textual review used in an experiment, and the right side suggests how they are labelled. + means the corresponding overall rating is considered as positive whilst − means the corresponding overall rating is considered as negative.

Table 2: Full sentiment analysis results

Method	Exp 1					Exp 2					Exp 3					Exp 4				
	Precision	Recall	F1-score	Accuracy	MCC	Precision	Recall	F1-score	Accuracy	MCC	Precision	Recall	F1-score	Accuracy	MCC	Precision	Recall	F1-score	Accuracy	MCC
<i>Lexicon-based Approaches</i>																				
LM	0.721	0.709	0.706	0.709	0.430	0.664	0.666	0.663	0.665	0.330	0.670	0.671	0.670	0.673	0.341	0.615	0.620	0.615	0.623	0.235
HIV4	0.694	0.648	0.626	0.648	0.339	0.594	0.544	0.495	0.584	0.129	0.579	0.530	0.464	0.573	0.096	0.551	0.519	0.466	0.601	0.063
SentiWordNet	0.738	0.715	0.707	0.715	0.453	0.636	0.623	0.622	0.639	0.258	0.651	0.633	0.631	0.652	0.284	0.614	0.599	0.600	0.637	0.213
VADER	0.782	0.768	0.765	0.768	0.550	0.764	0.694	0.690	0.721	0.453	0.756	0.669	0.658	0.700	0.416	0.712	0.621	0.610	0.686	0.320
<i>Machine Learning Approaches</i>																				
BOW + LR	0.861	0.861	0.861	0.861	0.722	0.859	0.851	0.853	0.857	0.709	0.868	0.861	0.864	0.867	0.730	0.798	0.778	0.784	0.800	0.575
BOW + NB	0.857	0.857	0.857	0.857	0.714	0.842	0.834	0.836	0.840	0.676	0.849	0.836	0.840	0.845	0.686	0.784	0.769	0.774	0.789	0.553
BOW + SVM	0.872	0.871	0.871	0.871	0.743	0.831	0.819	0.822	0.827	0.649	0.862	0.854	0.857	0.860	0.716	0.790	0.774	0.780	0.794	0.564
BOW + DT	0.797	0.786	0.784	0.786	0.582	0.749	0.739	0.741	0.749	0.488	0.780	0.757	0.760	0.771	0.536	0.701	0.673	0.677	0.709	0.373
BOW + RF	0.853	0.853	0.853	0.853	0.707	0.845	0.832	0.836	0.840	0.677	0.854	0.844	0.847	0.851	0.698	0.782	0.749	0.757	0.779	0.530
BOW + XGB	0.861	0.861	0.861	0.861	0.722	0.848	0.841	0.844	0.847	0.689	0.864	0.859	0.861	0.863	0.723	0.794	0.772	0.779	0.795	0.565
TF-IDF + LR	0.877	0.876	0.876	0.876	0.753	0.854	0.846	0.849	0.852	0.700	0.874	0.869	0.871	0.873	0.743	0.813	0.792	0.799	0.814	0.605
TF-IDF + NB	0.871	0.871	0.871	0.871	0.742	0.866	0.861	0.863	0.865	0.727	0.883	0.880	0.881	0.883	0.763	0.817	0.796	0.803	0.817	0.612
TF-IDF + SVM	0.877	0.877	0.877	0.877	0.753	0.844	0.824	0.829	0.835	0.668	0.872	0.859	0.863	0.867	0.731	0.818	0.747	0.758	0.789	0.560
TF-IDF + DT	0.800	0.789	0.787	0.789	0.590	0.758	0.742	0.745	0.754	0.500	0.760	0.747	0.749	0.757	0.506	0.679	0.666	0.669	0.694	0.345
TF-IDF + RF	0.855	0.855	0.855	0.855	0.711	0.842	0.817	0.822	0.829	0.659	0.858	0.842	0.846	0.851	0.700	0.775	0.726	0.735	0.764	0.499
TF-IDF + XGB	0.861	0.861	0.861	0.861	0.722	0.838	0.817	0.822	0.828	0.654	0.861	0.851	0.854	0.858	0.712	0.780	0.759	0.765	0.783	0.538
Word2Vec + LR	0.854	0.854	0.854	0.854	0.708	0.750	0.746	0.747	0.752	0.495	0.809	0.809	0.809	0.811	0.618	0.751	0.746	0.748	0.761	0.496
Word2Vec + NB	0.834	0.831	0.830	0.831	0.665	0.736	0.734	0.734	0.739	0.469	0.754	0.754	0.744	0.744	0.507	0.709	0.719	0.704	0.706	0.428
Word2Vec + SVM	0.853	0.852	0.852	0.852	0.705	0.752	0.752	0.752	0.755	0.504	0.813	0.815	0.814	0.816	0.628	0.755	0.753	0.754	0.765	0.507
Word2Vec + DT	0.841	0.841	0.841	0.841	0.682	0.752	0.747	0.749	0.754	0.499	0.801	0.804	0.801	0.803	0.604	0.735	0.730	0.732	0.746	0.465
Word2Vec + RF	0.861	0.861	0.861	0.861	0.722	0.756	0.753	0.754	0.759	0.509	0.826	0.825	0.825	0.828	0.651	0.762	0.755	0.758	0.771	0.517
Word2Vec + XGB	0.850	0.850	0.850	0.850	0.701	0.770	0.766	0.768	0.772	0.536	0.823	0.821	0.822	0.825	0.644	0.757	0.753	0.755	0.767	0.509
GloVe + LR	0.888	0.887	0.887	0.887	0.775	0.851	0.849	0.850	0.852	0.700	0.872	0.871	0.871	0.873	0.743	0.772	0.765	0.768	0.781	0.538
GloVe + NB	0.794	0.790	0.789	0.790	0.584	0.745	0.748	0.743	0.743	0.493	0.760	0.763	0.758	0.759	0.523	0.679	0.688	0.675	0.678	0.367
GloVe + SVM	0.887	0.886	0.886	0.886	0.773	0.850	0.850	0.850	0.852	0.701	0.860	0.861	0.860	0.862	0.721	0.775	0.770	0.772	0.784	0.545
GloVe + DT	0.778	0.777	0.777	0.777	0.555	0.734	0.733	0.734	0.738	0.468	0.748	0.750	0.749	0.751	0.498	0.697	0.693	0.695	0.711	0.390
GloVe + RF	0.869	0.867	0.867	0.867	0.736	0.849	0.833	0.837	0.842	0.682	0.857	0.839	0.844	0.849	0.696	0.778	0.746	0.754	0.776	0.523
GloVe + XGB	0.880	0.879	0.879	0.879	0.759	0.866	0.859	0.862	0.864	0.725	0.875	0.872	0.873	0.875	0.747	0.773	0.762	0.766	0.780	0.535
<i>Pre-trained Models</i>																				
BERT-base-cased	0.953	0.953	0.953	0.953	0.907	0.927	0.921	0.924	0.925	0.848	0.935	0.933	0.934	0.935	0.868	0.864	0.856	0.860	0.867	0.720
BERT-base-uncased	0.955	0.954	0.954	0.954	0.909	0.937	0.932	0.934	0.935	0.868	0.936	0.929	0.931	0.933	0.864	0.859	0.850	0.854	0.861	0.708
FinBERT	0.956	0.956	0.956	0.956	0.912	0.891	0.880	0.884	0.886	0.771	0.900	0.889	0.893	0.895	0.789	0.843	0.811	0.821	0.835	0.653

Table 2: Full sentiment analysis results

Method	Exp 5					Exp 6					Exp 7				
	Precision	Recall	F1-score	Accuracy	MCC	Precision	Recall	F1-score	Accuracy	MCC	Precision	Recall	F1-score	Accuracy	MCC
<i>Lexicon-based Approaches</i>															
LM	0.586	0.603	0.577	0.603	0.188	0.583	0.584	0.578	0.578	0.167	0.137	0.188	0.150	0.207	-0.007
HIV4	0.546	0.519	0.495	0.684	0.060	0.525	0.509	0.393	0.465	0.031	0.144	0.242	0.125	0.260	0.006
SentiWordNet	0.585	0.586	0.585	0.656	0.171	0.583	0.569	0.535	0.546	0.152	0.234	0.384	0.276	0.319	0.085
VADER	0.671	0.609	0.616	0.733	0.273	0.644	0.579	0.500	0.540	0.213	0.091	0.220	0.106	0.236	0.011
<i>Machine Learning Approaches</i>															
BOW + LR	0.733	0.691	0.704	0.776	0.422	0.709	0.711	0.709	0.711	0.419	0.532	0.533	0.529	0.573	0.331
BOW + NB	0.720	0.687	0.698	0.768	0.406	0.692	0.695	0.692	0.693	0.387	0.508	0.513	0.508	0.549	0.297
BOW + SVM	0.751	0.719	0.731	0.791	0.468	0.712	0.714	0.708	0.708	0.426	0.546	0.544	0.540	0.587	0.352
BOW + DT	0.696	0.598	0.601	0.740	0.277	0.629	0.630	0.629	0.634	0.259	0.471	0.463	0.423	0.529	0.250
BOW + RF	0.748	0.626	0.637	0.763	0.354	0.706	0.695	0.697	0.707	0.401	0.530	0.513	0.466	0.577	0.340
BOW + XGB	0.741	0.692	0.707	0.780	0.430	0.709	0.710	0.709	0.712	0.418	0.544	0.538	0.525	0.589	0.353
TF-IDF + LR	0.790	0.700	0.722	0.802	0.482	0.736	0.730	0.732	0.738	0.466	0.569	0.567	0.557	0.610	0.389
TF-IDF + NB	0.767	0.714	0.731	0.798	0.478	0.735	0.734	0.734	0.738	0.469	0.552	0.554	0.547	0.593	0.363
TF-IDF + SVM	0.813	0.573	0.554	0.747	0.301	0.728	0.697	0.698	0.716	0.424	0.550	0.505	0.444	0.582	0.361
TF-IDF + DT	0.684	0.606	0.612	0.738	0.279	0.617	0.618	0.617	0.619	0.235	0.481	0.466	0.427	0.521	0.242
TF-IDF + RF	0.780	0.606	0.608	0.760	0.344	0.699	0.689	0.691	0.701	0.388	0.490	0.504	0.442	0.576	0.343
TF-IDF + XGB	0.731	0.687	0.701	0.774	0.416	0.693	0.693	0.693	0.696	0.385	0.542	0.540	0.528	0.588	0.352
Word2Vec + LR	0.727	0.654	0.669	0.765	0.374	0.693	0.686	0.687	0.696	0.378	0.535	0.533	0.495	0.588	0.358
Word2Vec + NB	0.659	0.691	0.650	0.669	0.348	0.659	0.655	0.656	0.664	0.314	0.502	0.506	0.493	0.513	0.274
Word2Vec + SVM	0.735	0.661	0.677	0.770	0.389	0.690	0.683	0.685	0.693	0.374	0.719	0.523	0.444	0.585	0.363
Word2Vec + DT	0.707	0.640	0.652	0.754	0.340	0.663	0.650	0.650	0.665	0.313	0.503	0.514	0.483	0.561	0.315
Word2Vec + RF	0.727	0.661	0.676	0.767	0.383	0.679	0.677	0.678	0.684	0.357	0.505	0.517	0.489	0.570	0.325
Word2Vec + XGB	0.735	0.694	0.708	0.778	0.427	0.686	0.684	0.685	0.690	0.370	0.519	0.526	0.515	0.569	0.324
GloVe + LR	0.747	0.698	0.713	0.785	0.442	0.705	0.699	0.700	0.708	0.404	0.542	0.548	0.525	0.596	0.369
GloVe + NB	0.637	0.666	0.627	0.647	0.302	0.628	0.618	0.617	0.633	0.245	0.476	0.487	0.469	0.497	0.244
GloVe + SVM	0.751	0.701	0.717	0.787	0.449	0.705	0.698	0.699	0.707	0.402	0.514	0.536	0.499	0.589	0.360
GloVe + DT	0.674	0.631	0.641	0.736	0.301	0.619	0.615	0.615	0.625	0.234	0.338	0.460	0.389	0.515	0.239
GloVe + RF	0.758	0.642	0.656	0.771	0.382	0.676	0.673	0.674	0.681	0.349	0.512	0.507	0.472	0.569	0.322
GloVe + XGB	0.734	0.690	0.704	0.777	0.422	0.690	0.687	0.688	0.694	0.377	0.543	0.547	0.534	0.591	0.359
<i>Pre-trained Models</i>															
BERT-base-cased	<u>0.767</u>	<u>0.786</u>	<u>0.775</u>	<u>0.807</u>	<u>0.553</u>	<u>0.771</u>	<u>0.774</u>	<u>0.770</u>	<u>0.771</u>	<u>0.545</u>	0.624	0.614	0.617	0.629	0.437
BERT-base-uncased	0.758	0.780	0.767	0.799	0.538	0.765	0.765	0.765	0.767	0.530	0.643	0.632	0.635	0.650	0.466
FinBERT	0.777	0.718	0.737	0.803	0.492	0.743	0.746	0.741	0.742	0.488	0.605	0.579	0.585	0.618	0.405

Notes: This table summarises the sentiment analysis results for all 7 experiments. The first 6 experiments are binary sentiment classifications, and the last one is a three-class sentiment classification introducing the neutral sentiment. The best result of each method category is in bold, and the best one of each experiment is underscored.

Appendix 1: Literature review summary

Table 3

Paper	Text	Label	LX	WE	ML	PM	Findings
Antweiler and Frank (2004)	Yahoo! Finance, Raging Bull message boards	Manually Annotated	-	BOW	NB, SVM	-	The effect of messages on stock returns is statistically significant but economically small. NB and SVM produce similar sentiment classification results.
Tetlock et al. (2008)	Wall Street Journal, Dow Jones News	Not Required	HIV4	-	-	-	Negative words in financial news stories forecast low firm earnings. Firms' stock prices reflect information embedded in negative terms with a slight delay. Negative words related to fundamentals have the most significant impact on earnings and return predictability.
Kothari et al. (2009)	Dow Jones News, Investext, Factiva, SEC EDGAR	Not Required	HIV4	-	-	-	Negative disclosures from business press sources drive up the cost of capital and return volatility.
Li (2010)	10-Ks and 10-Qs	Manually annotated	HIV4, LIWC, DICTION	BOW	NB	-	The tone of the forward-looking statements is positively associated with future earnings. The dictionary-based sentiment measures fail to predict future performance.
Feldman et al. (2010)	10-Ks and 10-Qs	Not Required	HIV4, LM	-	-	-	The sentiment change in the MD&A section of the SEC filing is a significant predictor of short-term market reactions.
Rogers et al. (2011)	Corporate Disclosures	Not Required	DICTION, LM	-	-	-	Firms with more optimistic statements in their earnings announcements suffer from higher litigation risk.
Loughran and McDonald (2011)	10-Ks	Not Required	LM	-	-	-	Creation of an alternative negative word list that suits financial text better than Harvard IV-4.
Engelberg et al. (2012)	Dow Jones News	Not Required	HIV4, LM	-	-	-	The negative relation between short sales and future returns is more severe in negative news.
Hagenau et al. (2013)	German Adhoc, EuroAdhoc	Market Reaction	-	TF-IDF	SVM	-	Feature selection with bi-grams significantly improves sentiment classification accuracies of financial news, which help to improve stock price prediction.

Table 3

Paper	Text	Label	LX	WE	ML	PM	Findings
Huang et al. (2014)	Investext	Manually Annotated	HIV4, LM, LIWC, DICTION	BOW	NB	-	Investors react more strongly to negative texts than to positive ones. NB is more effective in extracting opinions from analyst reports than dictionary-based approaches.
Tsai and Wang (2017)	10-Ks	Not Required	LM	-	-	-	Using regression and ranking methods, the experimental results show that soft information such as finance-specific sentiment lexicon strongly correlates to financial risks.
Sohangir et al. (2018)	StockTwits	Self-reported	VADER, WordNet	N-grams	LR, NB, SVM	-	VADER outperforms machine learning methods in extracting sentiment from financial social media.
Sousa et al. (2019)	Financial News	Manually Annotated	-	BOW, TF-IDF	NB, SVM, TextCNN	BERT	Sentiment change in financial news achieved a 69% hit rate in predicting stock exchange variation. BERT has the highest accuracy of 82.5% among the sentiment analysis methods.
Araci (2019)	Financial Phrase Bank, FiQA	Manually Annotated	-	-	-	BERT, FinBERT	Further pre-training BERT on a subset of Reuters' TRC2 1.8M news articles, their vision of FinBERT improved BERT by 15% in accuracy.
Feuerriegel and Gordon (2019)	German Adhoc	Not Required	LM	-	-	-	High-dimensionality text input from financial news overfits machine learning models when predicting macroeconomic indicators; feature reduction can be achieved by mapping semantic categories onto latent structures.
Abbasi et al. (2019)	Twitter, Forum Channels	Manually Annotated		N-grams	RF, LR, SVM	-	User-generated content on social media can detect adverse events in advance, and false-positive rates are further reduced after including negative sentiment polarity in the models.
Renault (2020)	StockTwits	Self-reported	-	N-grams	MaxEnt, MLP, NB, RF, SVC	-	MaxEnt has the highest accuracy of 74.451% in sentiment analysis, followed by SVM with an accuracy of 74.292%. There is no substantial evidence that investors' opinion on social media helps predict significant capitalization stock returns daily.

Table 3

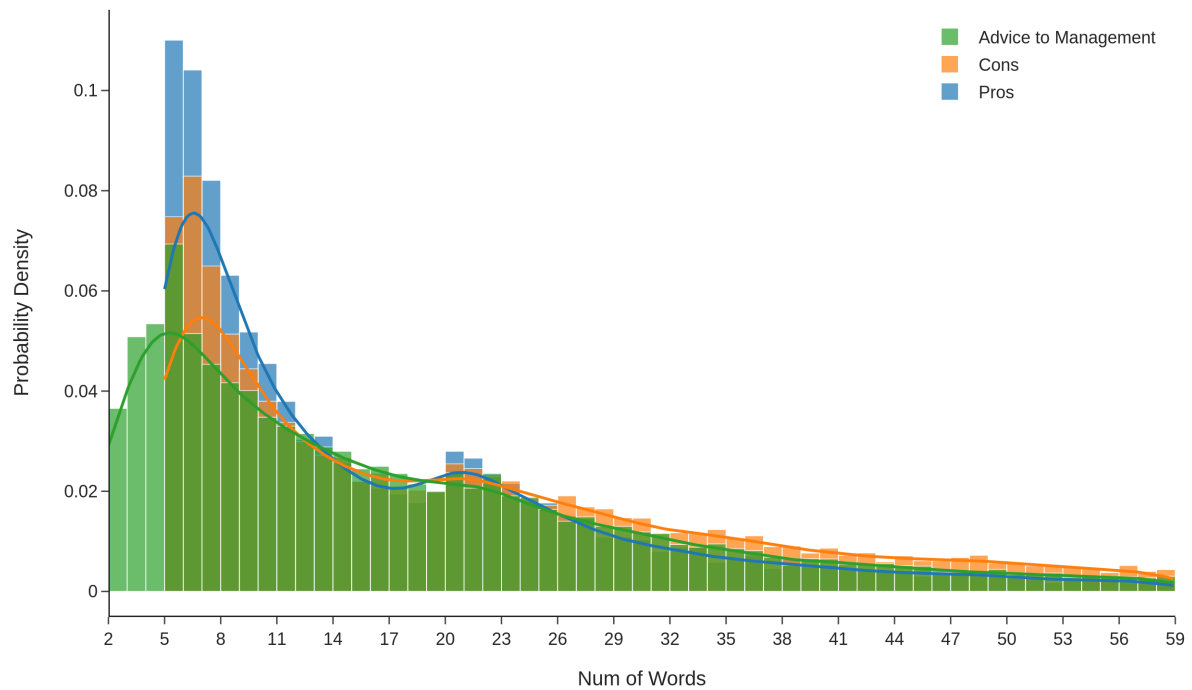
Paper	Text	Label	LX	WE	ML	PM	Findings
Yang et al. (2020)	Financial Phrase Bank, Analyst Tone, FiQA	Manually Annotated	-	-	-	BERT, FinBERT	Pre-training BERT-like on a sizeable financial corpus with 4.9 billion tokens (e.g. 10-Ks and 10-Qs, earnings call transcripts and analyst reports), FinBERT demonstrates superiority in handling financial sentiment classification tasks.
Mishev et al. (2020)	Financial Phrase Bank, SemEval 2017 TASK 5	Manually Annotated	HIV4, LM	BOW, TF-IDF, Word2Vec, FastText, GloVe, ELMO	Attention, BiGRU, BiLSTM, CNN, Dense, LSTM, SVC, XGB	BERT, XLNet, XLM, FinBERT, DistilBERT, ALBERT, RoBERTa	The BERT models show better performances than machine learning methods, mainly because semantic meaning enriches text representation.
Stevenson et al. (2021)	Anonymous Credit Lender	Default Record	-	TF-IDF	LR, RF	BERT	BERT outperformed LR and RF in predicting small business loan default. Textual loan information produces relatively accurate predictions alone. However, it does not offer additional performance lift when used with structured data.
Jaggi et al. (2021)	Stocktwits	Market Reaction	-	BOW, TF-IDF	LR, NB, RF, XGB	BERT, FinBERT, ALBERT, FinALBERT	The best results are given by the BERT and NB across one year and two years of data for binary and 3-class sentiment classification.
Kriebel and Stitz (2021)	Lending Club	Default Record	-	TF-IDF, GloVe	CNN, RNN, CRNN, AE	BERT, RoBERTa	Textual information can improve credit default predictions. While machine learning models hold good results, BERT models have better performance in nearly all cases.
Frankel et al. (2022)	10-Ks, Earnings Call Transcripts	Market Reaction	HIV4, LM	N-grams	RF, sLDA, SVM	-	Machine learning methods are not only implementable and reliable measures of disclosure sentiment but also more potent than dictionary-based methods. RF is better at capturing disclosure sentiment than other machine learning methods.
Huang et al. (2022)	Analyst Reports	Manually Annotated	LM	BOW, N-grams	NB, SVM, RF, CNN, LSTM	FinBERT	FinBERT incorporates finance knowledge and can better summarise contextual information in financial texts.

Notes: LX: Lexicon-based Approach, WE: Word Embedding, ML: Machine Learning Approach, PM: Pre-trained Model.

Appendix 2: Glassdoor employee review

Reviewers have to provide text comments with a minimum of five words on *pros* and *cons* for the company. There is an optional text field to provide *advice to management*. From the distribution in Panel A we can see the majority of reviews from *pros* and *cons* in our sample are between 5-8 words, but the probability of *pros* is higher than *cons* and *advice to management*. Comments with more than 23 words have a higher chance of coming from the *cons*. Panel B provides a visual representation of words in our sample data.

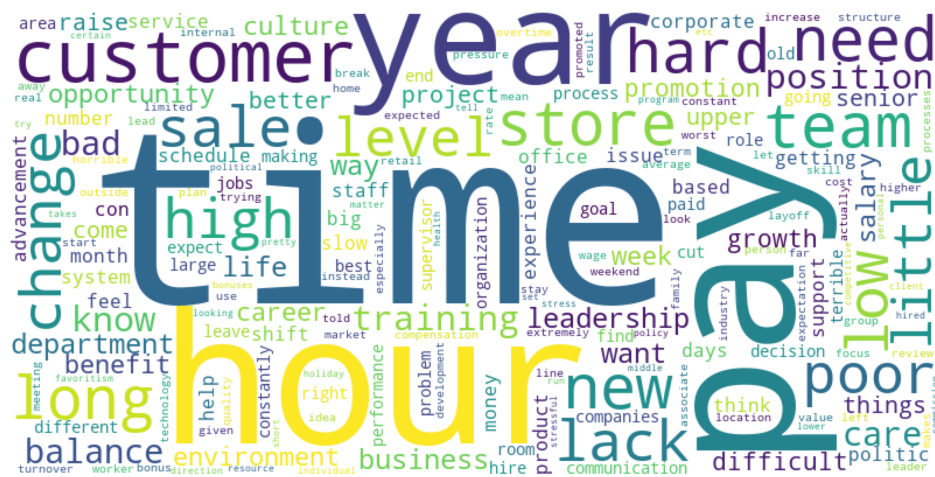
Panel A: Distribution of review lengths in *pros*, *cons* and *advice to management*.



Panel B: Word cloud of *pros*, *cons* and *advice to management*.



(a) Pros



(b) Cons



(c) Advice to Management

Appendix 3: Models description

Method	Model Description
<i>Lexicon-based Approaches</i>	
LM	Loughran-McDonald features + Lydia sentiment analysis system
HIV4	Harvard IV-4 dictionary features + Lydia sentiment analysis system
SentiWordNet	SentiWordNet features + Sentiment polarity
VADER	Compound valence score
<i>Machine Learning Approaches</i>	
BOW + LR	Count Vectorizer + Logistic Regression (C = 10)
BOW + NB	Count Vectorizer + Multinomial NB
BOW + SVM	Count Vectorizer + SVC (kernel = "linear", C = 1)
BOW + DT	Count Vectorizer + DT classifier (max_depth = 11)
BOW + RF	Count Vectorizer + RF classifier (n_estimators = 100)
BOW + XGB	Count Vectorizer + XGB classifier (learning_rate = 0.3, max_depth = 8)
TF-IDF + LR	TF-IDF Vectorizer + Logistic Regression (C = 10)
TF-IDF + NB	TF-IDF Vectorizer + Multinomial NB
TF-IDF + SVM	TF-IDF Vectorizer + SVC (kernel = "linear", C = 1)
TF-IDF + DT	TF-IDF Vectorizer + DT classifier (max_depth = 7)
TF-IDF + RF	TF-IDF Vectorizer + RF classifier (n_estimators = 100)
TF-IDF + XGB	TF-IDF Vectorizer + XGB classifier (learning_rate=0.3, max_depth = 10)
Word2Vec + LR	Word2Vec (average) + Logistic Regression (C = 10)
Word2Vec + NB	Word2Vec (average) + Multinomial NB
Word2Vec + SVM	Word2Vec (average) + SVC (kernel = "linear", C = 1)
Word2Vec + DT	Word2Vec (average) + DT classifier (max_depth = 4)
Word2Vec + RF	Word2Vec (average) + RF classifier (n_estimators = 210)
Word2Vec + XGB	Word2Vec (average) + XGB classifier (learning_rate = 0.3, max_depth = 10)
GloVe + LR	GloVe (average) + Logistic Regression (C = 10)
GloVe + NB	GloVe (average) + Multinomial NB
GloVe + SVM	GloVe (average) + SVC (kernel = "linear", C = 1)
GloVe + DT	GloVe (average) + DT classifier (max_depth = 6)
GloVe + RF	GloVe (average) + RF classifier (n_estimators = 190)
GloVe + XGB	GloVe (average) + XGB classifier (learning_rate = 0.3, max_depth = 6)
<i>Pre-trained Models</i>	
BERT-base-cased	110M parameters, pre-trained on cased Wikipedia and BookCorpus, AdamW: learning rate = 2e-5, batch size = 32, epoch = 5
BERT-base-uncased	110M parameters, pre-trained on lower-cased Wikipedia and BookCorpus, AdamW: learning rate = 2e-5, batch size = 32, epoch = 5
FinBERT	110M parameters, pre-train BERT on financial communication text, AdamW: learning rate = 2e-5, batch size = 32, epoch = 5

Appendix 4: Performance of untuned DT

Decision trees have the lowest performance of all machine learning models, mainly because they are too easy to overfit the data. During training, a decision tree evaluates all possible splits and grows iteratively until it reaches the terminating nodes. Tuning the maximum depth in this case helps to reduce overfitting by limiting the expansion of the tree. Initially, we did not specify the value of maximum depth and found VADER outperform DT in most cases of the binary sentiment analysis, this proves that although the dictionary approaches are weak, they have the potential to improve if the dictionaries can be tuned subject to the task as well.

Method	Exp 1					Exp 2				
	Precision	Recall	F1-score	Accuracy	MCC	Precision	Recall	F1-score	Accuracy	MCC
VADER	0.782	0.768	0.765	0.768	0.550	0.764	0.694	0.690	0.721	0.453
BOW + DT	0.775	0.775	0.775	0.775	0.549	0.729	0.722	0.724	0.731	0.451
TF-IDF + DT	0.783	0.782	0.782	0.782	0.565	0.718	0.713	0.715	0.721	0.431
Word2Vec + DT	0.804	0.804	0.804	0.804	0.609	0.663	0.664	0.663	0.667	0.327
GloVe + DT	0.745	0.745	0.745	0.745	0.491	0.700	0.698	0.699	0.703	0.398

Method	Exp 3					Exp 4				
	Precision	Recall	F1-score	Accuracy	MCC	Precision	Recall	F1-score	Accuracy	MCC
VADER	0.756	0.669	0.658	0.700	0.416	0.712	0.621	0.610	0.686	0.320
BOW + DT	0.753	0.744	0.746	0.753	0.497	0.665	0.664	0.664	0.680	0.328
TF-IDF + DT	0.711	0.704	0.705	0.713	0.415	0.652	0.648	0.650	0.669	0.300
Word2Vec + DT	0.740	0.738	0.739	0.743	0.478	0.680	0.680	0.680	0.694	0.360
GloVe + DT	0.715	0.716	0.715	0.718	0.431	0.638	0.640	0.639	0.653	0.278

Method	Exp 5					Exp 6				
	Precision	Recall	F1-score	Accuracy	MCC	Precision	Recall	F1-score	Accuracy	MCC
VADER	0.671	0.609	0.616	0.733	0.273	0.644	0.579	0.500	0.540	0.213
BOW + DT	0.628	0.617	0.621	0.699	0.244	0.596	0.597	0.597	0.600	0.193
TF-IDF + DT	0.599	0.592	0.594	0.677	0.191	0.591	0.592	0.591	0.593	0.184
Word2Vec + DT	0.627	0.630	0.628	0.689	0.256	0.601	0.601	0.601	0.605	0.202
GloVe + DT	0.578	0.583	0.579	0.640	0.160	0.580	0.580	0.580	0.584	0.159

Method	Exp 7				
	Precision	Recall	F1-score	Accuracy	MCC
VADER	0.091	0.220	0.106	0.236	0.011
BOW + DT	0.446	0.443	0.443	0.472	0.178
TF-IDF + DT	0.440	0.440	0.440	0.464	0.170
Word2Vec + DT	0.443	0.441	0.442	0.461	0.172
GloVe + DT	0.425	0.424	0.425	0.441	0.141

Appendix 5: Robustness check

The figures below show the change in the mean values of the other evaluation metrics when text reviews are associated with star ratings over different intervals. The overall results are the same as for our main evaluation metric MCC, except that the mean recall for the BERT model and the mean recall and F1 scores for the machine learning approach is higher in Experiment 6 than in Experiment 5. This is because, in Experiment 5, the classes are skewed to positive sentiment whereas, in Experiment 6, the classes of positive and negative sentiment are more balanced. Researchers should cautiously choose the evaluation metrics when dealing with unbalanced data.

